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Pair trading strategies in the cryptoassets market: a cointegration framework with optimized thresholds using genetic algorithms

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The crypto-assets markets are notoriously volatile and risky. In this context, market-neutral type strategies, such as pair-trading, may be relevant. In this paper, we focus on the implementation of pair-trading strategies with a wide range of crypto-assets over periods between August 2021 and January 2024. To carry out this study, we combine econometric and machine learning techniques which differ from those used in existing literature on the subject. By using cointegration tests and error correction models, we identify a sample of 229 pairs suitable for pair-trading strategies. Using a genetic algorithm and pair clustering, we test four strategies using standard and optimized thresholds. The results highlight the existence of profitable cointegrating relationships, and, therefore, short-term market inefficiencies in the crypto-assets market. Indeed, though still risky, the best strategy identified in terms of risk-return trade-off, with a median maxdrawdown of 15.29%, delivers an average annual Sharpe ratio per pair of 0.69 over the out-of-sample period.

Keywords: Cryptoassets; Pair-trading; Cointegration; Error-correction models; Genetic algorithm; K-means; Short-term market inefficiencies

JEL Classifications: C22, C61, G11, G12, G14

1. Introduction

Since their emergence in 2008 with the creation of the Bitcoin blockchain, digital assets have enjoyed a spectacular rise in popularity. This boom, bringing the overall valuation of this emerging asset class to $1,660 billion, as of January 31, 2024, is undoubtedly due to two factors.

On the one hand, the promise of digital assets, and Bitcoin in particular, to break away from the traditional financial system and create a peer-to-peer system has found its audience against the backdrop of the economic crisis in 2008 and, more recently, the health crisis in 2020.

On the other hand, the technological innovation of blockchain, which is the cornerstone of this new asset class, makes digital assets particularly attractive. The growing interest in digital assets is driving a significant increase in transaction volumes, creating a dynamic market environment. This new demand opens up new perspectives for traditional investment strategies. Due to their high volatility, these assets offer interesting arbitrage opportunities to enhance portfolio performance by adapting traditional or market-neutral investment strategies, such as pair-trading.

Pair-trading is a statistical arbitrage strategy. It takes advantage of the temporary divergence between the prices, also known as the ‘spread’, of two assets linked over the long term. When such a divergence occurs, the overvalued asset is sold short and the undervalued asset is bought. Positions are closed when the divergence disappears. The profit on this strategy is the sum of the gains on the long and short positions. Since its inception in the 1980s, pair-trading has been a popular strategy for investment banks and hedge funds, as it allows them to avoid exposure to the market. In other words, profits are independent of market trends. This type of strategy offers two advantages in a financial market context.

(1) Market-neutral type strategies, which include pair-trading, are known to adapt particularly well to turbulent market situations.

(2) The theory of market efficiency, developed by Fama (1970), asserts that asset prices adjust instantaneously to their fundamental values based on available information,

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making prices perfectly unpredictable. It is therefore impossible to predict returns higher than the average market return. However, in the context of a pair-trading strategy, this prediction is modified because: (i) it does not relate to the individual prices of an asset but to a linear combination of the prices of two assets; (ii) it is not systematic at every period (day, hour…) but punctual when the deviation of the price combination from its historical average is significant. This implies that if these arbitrage opportunities exist, they could illustrate short-term inefficiencies in digital asset markets.

While pair-trading is widespread on traditional financial markets, it is still under-exploited on the crypto-asset market. Although some studies, such as those by Fil and Kristoufek (2020), Lesa and Hochreiter (2023) and Nair (2021), have already explored this topic, their work has often focused on a limited number of cryptocurrencies and the application of various econometric methods. This paper aims to extend this research by examining a broad range of cryptocurrencies and integrating econometric methods, specifically cointegration, with machine learning through a genetic algorithm to optimize the thresholds of our strategies. Therefore, our study aims to answer the following question: Are pair-trading strategies applicable and profitable in the digital asset market? If so, what implications do they have for underlying fundamental value dynamics and market efficiency?

To answer this question, we focus on pair-trading strategies within a cointegration framework. The notion of cointegration reflects the principle of a common stochastic trend between crypto-asset prices. It is conceivable that this common trend could be influenced by news affecting not only investor confidence (distrust), but also the business models of the projects and companies developing these crypto-assets, considering them as both payment instruments and investment vehicles. Furthermore, it is important to note that the investors' profile in traditional markets differs significantly from that of crypto-asset investors. Traditional markets mainly attract institutional investors and well-informed individuals, whereas crypto-asset market includes a significant proportion of younger and sometimes less-experienced retail investors.† If pair-trading strategies prove successful in the crypto-asset market, this would indicate a certain rationality in investment behavior and suggest that crypto-assets possess fundamental values recognized by investors due to a return mechanisms system for every deviation. This could demonstrate that, despite the volatility and specificities of this market, rational and profitable arbitrage opportunities exist. These arbitrage opportunities would be the revelation of temporary pockets of inefficiency in the market which, over the long term, display price spreads compatible with the fundamental value of assets.

Across a wide range of digital assets, we identify and optimize the entry and exit levels of positions in several strategies on the selected asset pairs. To be more precise, the technical elements of our approach are as follows:

Using Engle and Granger (1987) tests, we examine cointegration relationships between asset prices. Our sample consists of 208 crypto-assets listed on the Binance exchange platform. To obtain an optimal strategy, we go through several steps: (i) we use all the information provided by error-correction models, enabling us, among other things, to identify the assets participating in the return to equilibrium of the long-term relationships of each pair. In particular, we exclude pairs where one of the two cryptocurrencies does not participate in the return to equilibrium (i.e. weak exogeneity). (ii) We compare four main strategies: fixed strategy, dynamic strategy, percentage strategy and return strategy to identify the best in terms of risk/return trade-off. (iii) Finally, we optimize strategy parameters using a genetic algorithm to extract the full potential of pair-trading strategies in this market. Our sample is divided into two distinct periods. The pair identification and strategy optimization period runs from August 1, 2021 to July 31, 2023, while the backtesting period runs from August 1, 2023 to January 31, 2024.

Our results suggest the existence of cointegrating relationships between two assets on the crypto-asset market. Indeed, the various cointegration tests carried out agree on the presence of 1541 cointegrated pairs between 162 assets. Although this figure is high, not all the identified pairs are compatible with a pair-trading strategy. After applying several filters to select only the most relevant pairs, we are able to isolate 229 pairs. A comparison between the first two strategies identified, namely the fixed and dynamic strategies, leads to the identification of changes in the volatility of the pairs' long-term relationships. Indeed, with the first one, only 28% of pairs take at least one position over the backtesting period. This can be explained by the fact that, for the majority of pairs, the residual of their cointegrating relationship does not reach the trigger thresholds, defined as a multiple of the standard deviation of this same residual over the pair identification period.

The optimization tests finally yield an approach delivering a median Sharpe ratio on the 229 pairs of 0.61 for an average max drawdown on each pair of -20.79% over the backtesting period. The profits delivered by the strategies clearly suggest the existence of pockets of market inefficiency, associated with temporary anomalies in asset price dynamics.

The rest of this paper is organized as follows.

Section 2 is devoted to literature reviews. Section 3 presents the empirical strategy of our paper, introducing the data sample used, and illustrating the principles of pair selection through cointegration tests. We introduce an iterative methodology, incorporating error-correction models for each pair. At the end of this section, we illustrate the methodology to test the different pair-trading approaches. The results of the different strategies applied to the out-of-sample phase are presented in Section 4. Section 5 summarizes our main results, analyzing and discussing them in depth, particularly in relation to their implications for crypto-asset market efficiency. Finally, Section 6 summarizes our study and suggests potential improvements for further research.

† From Retail to Institutional: The Shift in Crypto Investment. https://www.benzinga.com/markets/cryptocurrency/23/02/30905456/from-retail-to-institutional-the-shift-in-crypto-investment (accessed 1 September 2024). Spot ETFs in Crypto Markets. https://www.binance.com/en/research/analysis/spot-etfs-in-crypto-markets/ (accessed 1 September 2024).

2. Literature review

Amongst the literature already published, some papers have already centered on pair-trading strategies on the cryptocurrency sector but, often, focusing on a limited number of assets. Figá-Talamanca *et al.* (2021) focus in their study on four cryptocurrencies characterizing the market: Bitcoin, Ethereum, Litecoin and Monero. Singh *et al.* (2023) focus on two cryptocurrencies (Ethereum and Bitcoin) to compare two trading strategies (double momentum and pair-trading). For their part, Leung and Nguyen (2019) construct cointegrated portfolios involving four cryptocurrencies: Bitcoin, Ethereum, Bitcoin Cash and Litecoin. Our study aims to break away from common a priori and beliefs by taking a large panel of digital assets in order to validate or invalidate these beliefs a posteriori.

The temporality of the data used in this type of study is also strategic and requires special consideration. While some articles study a specific temporality, others compare the results observed. Leung and Nguyen (2019) use the closing prices of a selection of cryptocurrencies on a daily basis. Similarly, Singh *et al.* (2023) compare two strategies using daily data, and their results are conditional on this temporality. On the other hand, Fil and Kristoufek (2020) compare results over several timeframes. In their study, they compare two pairs selection methods for applying pair-trading strategies to 26 cryptocurrencies. They use frequencies of five minutes, one hour and one day. While gross returns are often better in a high-frequency setting, the introduction of external factors (i.e. transaction fees) has a significant impact on intraday performance compared to a daily setting. Ko *et al.* (2023) look at different temporalities in a high-frequency setting (one minute, five minutes and one hour) to compare several pairs selection methods. The observed results indicate that performance is significantly better at high frequencies, suggesting that there is an intraday mean-reverting behavior that would not be replicated at the daily level. Although gross performance appears to be better in an intraday setting, taking into account the inclusion of external factors such as transaction costs or market liquidity leads us, in this study, to focus on a daily setting.

Implementing a pair-trading strategy involves two steps. The first is to select pairs with specific characteristics that provide adequate suitability to generate profits. The second step is to optimize the strategy with the identified pairs. The aim is to find entry and exit thresholds which generate the best return while controlling the investor's risk.

Pairs selection

Studies on the subject have developed a number of methods for selecting pairs when implementing a pair-trading strategy. The main two being distance and cointegration. Distance, the oldest, is a strategy that does not take account of economic factors, resulting in a lack of forecasting capacity. It consists in using the Euclidean quadratic distance on normalized price time series. The pairs with the smallest distance are selected to implement the strategy. The pioneers in the use of this method are Gatev *et al.* (2006). Studying traditional markets, the authors found that this type of method generated profits when pair-trading strategies were implemented on the selected pairs.

Cointegration refers to the phenomenon whereby two assets show similar price trends over the long term. Vidyamurthy (2004) provides a comprehensive guide to implementing cointegration in a pair-trading strategy, taking into account risks, transaction costs, and portfolio construction. Like distance, cointegration can present certain limitations, since it can be subject to structural breaks. Despite this, it remains a popular and widely-used method for identifying pairs within a pair-trading strategy.

In the traditional market, Huck and Afawubo (2015) investigates whether the cointegration method provide better results than the distance method. Their study, based on a basket of S&P 500 stocks, suggests that, after controlling for risk, the distance method delivers only insignificant excess returns. In contrast, cointegration generates robust, high and stable returns over time. Similar studies on the cryptocurrency market have also already been carried out. Fil and Kristoufek (2020) compared these two methods. Their results suggest that the performance of the distance method increases significantly with higher frequency. Conversely, the cointegration method appears to perform much better in the daily setting, but does not scale as well with frequency, with performance deteriorating in the hourly setting. Lesa and Hochreiter (2023) test these two methods on a basket of 20 cryptocurrencies with hourly and daily frequency. The results are similar to previous papers: in terms of temporality, intraday trading is more profitable in terms of gains with both methods when no external factors such as transaction fees or stop-loss are introduced. The cointegration method tends to outperform the distance method across the range of strategies and frequencies chosen.

Four other methods are illustrated in the literature. The correlation method calculates the correlation coefficient between two assets. This refers to the phenomenon whereby two assets show similar price trends over the short term. A high correlation indicates potentially profitable trading opportunities. An early work by Brooks *et al.* (2001) examined the lead-lag relationship between the FTSE 100 index and the futures price index, calculating the correlation and using the results to create a trading strategy. Although lagged changes in futures prices could predict changes in spot prices, the strategy was unable to beat the benchmark due to transaction costs.

The stochastic differential residuals method on asset returns, developed by Do *et al.* (2006), uses the Capital Asset Pricing Model (CAPM) and Arbitrage Pricing Theory (APT) to define the residual deviation function. This function, which considers the relative valuation of the two values as an equilibrium, represents the state of imbalance or mispricing. It is this function that captures any excess deviation from the spread value over the long term (which ultimately provides trading signals on the pair of two assets).

Recently, Stübinger and Bredthauer (2017) introduced the fluctuation behavior method. This involves identifying pairs with rapid mean reversion, measured by the number of times the pairs' spread crosses back through zero, and high volatility, measured by the standard deviation of the spread. The pairs with the highest combined ranks are selected.

Finally, the Hurst exponent method, used for the first time for pair selection in the article by Ramos-Requena *et al.* (2017), selects pairs with an exponent between 0 and

0.5, i.e. pairs exhibiting mean reversion or anti-persistent behavior.

All these methods were first implemented on traditional financial markets. Tourin and Yan (2013) used the stochastic control approach to construct pair-trading portfolios dynamically. Rad *et al.* (2016) carried out a study of the performance of three different pair-trading strategies: distance, cointegration and copula methods, on the US stock market from 1962 to 2014. While they have found that, in terms of continuity, copula methods are better able to adapt to financial crises, all three strategies deliver a positive average return. They perform better during volatile periods, but the cointegration strategy is superior during turbulent market conditions. Nair (2021) uses three of the methods mentioned (distance, cointegration and correlation) with four cryptocurrencies (Bitcoin, Ethereum, Litecoin, Neocoin) across four samples. The study shows that there is long-term convergence in cryptocurrency prices: when two prices are correlated in one sample, there is a high probability that they will be correlated in the next sample. All the strategies used in his study pay off (especially for pairs made up of large cryptocurrencies). Conversely, Ko *et al.* (2023) compare the observed results of pair-trading strategies over the six methods listed above. In the end, the distance method brings better results in terms of return parameters for all frequencies. In contrast, the cointegration method presents a lower risk for high frequencies.

Although distance sometimes provides superior returns, Krauss (2017) shows that the cointegration method provides a more rigorous framework than distance due to the econometric identification of relationships between assets. Furthermore, Keilbar and Zhang (2021) demonstrate that cryptocurrencies are cointegrated with a rank of four, indicating a long-run equilibrium relationship that can be exploited to develop strategies. In this paper, we therefore give preference to the cointegration method for selecting pairs, especially as this method is better in turbulent markets.

Strategy optimization

Once the pairs have been selected using one of the methods described above, indicators need to be set up to trigger buy and sell orders for a pair. In a pair-trading strategy, a buy (sell) order is triggered when the spread between the two assets falls below (above) a certain threshold. Several tactics exist for setting up the different thresholds. Huang and Martin (2019) compare three of them: a percentage method,† one with a fixed standard deviation‡ and one reminiscent of Bollinger band strategies.§

Once the strategies have been defined, their parameters need to be optimized. While Huang and Martin (2019) use two traditional optimization criteria, absolute profit and profit factor, there are now machine learning techniques whose aim is to optimize a predefined function: the genetic algorithm.

Developed by Holland (1992), genetic algorithm is an optimization technique inspired by the theory of evolution. It uses selection, crossover and mutation processes to find efficient solutions to complex problems. In recent years, it has been widely used in the field of finance to optimize investment strategies for stock market assets. The paper by Huang *et al.* (2015) uses Taiwan Stock Exchange shares to test the application of a genetic algorithm in the optimization of a pair-trading strategy. The algorithm optimizes the Bollinger bands used, the moving average and the weights of each asset within the strategy. Using this method, they show that models based on genetic algorithms significantly outperform a benchmark and produce robust models. Chen *et al.* (2022) use a genetic algorithm to incorporate more criteria in the implementation and optimization of a pair-trading strategy based on 44 stocks in the Taiwan 50 Index. The entire method, including pair selection, is encoded within the genetic algorithm. Finally, Sermpinis *et al.* (2020) compare three strategies: fixed thresholds defined as the benchmark strategy, a genetic algorithm to optimize the parameters of the Bollinger criteria, and a new structure for deep reinforcement learning. The study covers 35 commodities from 1980 to 2018. The results show insignificant returns for the first strategy, while the strategy using the genetic algorithm improves returns but increases the associated risk.

3. Modeling strategy

Compared with previous literature, our paper presents several original features.

Firstly, our study uses a large number of digital assets to identify pairs that are relevant for trading strategies. This research is carried out without any preconceived ideas about potential relationships between assets.

Secondly, our study is based on a longer timeframe than those usually used in existing literature. As this market is still in full growth, with new crypto-assets appearing on a regular basis, studies already carried out were unable to exploit a time period as long as ours.

Another unique aspect of our research is the focus on studying the behavior of each individual asset within a pair. Thanks to the construction of simple error correction models (ECM) associated with the cointegration relationship between the prices of the two assets, we are able to explain the mechanisms of return to equilibrium and identify whether each asset participates, or not, in the return to equilibrium. In this version of the paper, we focus on pairs where the return to equilibrium is achieved by both assets. This question touches on the hypothesis of the weak exogeneity of certain assets involved in cointegrating relationships, and also raises the question of which cryptoassets are leaders or followers in the price evolution dynamics.

Finally, we use a genetic algorithm to optimize the entry and exit thresholds in our strategies. This innovative approach combines econometrics with cointegration, error-correction models for pair selection and machine learning for strategy optimization.

† It sets the minimum residual of the cointegration relationship at 0% and the maximum at 100%. It has six levels: sell-entry, sell-stop-loss, sell-take-profit, buy-entry, buy-stop-loss and buy-take-profit, each corresponding to a certain percentage that must be optimized.

‡ Similar to percentage, except that here it uses different multiples of the long-term standard deviation to replace percentage levels, with the mean of the cointegrating residuals being zero.

§ The difference with the long-term standard deviation strategy is that here standard deviation levels are no longer fixed, but dynamically readjusted.

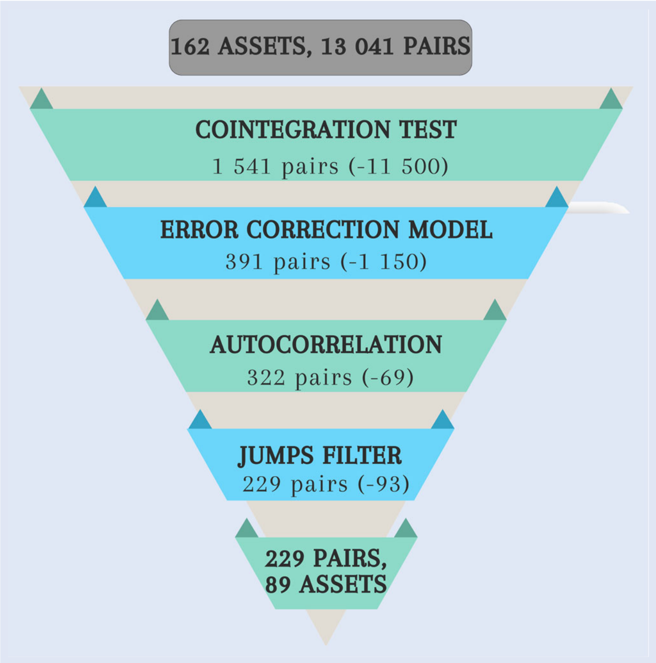


Figure 1. Iterative pair selection method.

3.1. *Data*

Data comes from the Binance exchange platform. We use the closing prices of 208 digital assets,¶ retrieved daily from August 1, 2021 to January 31, 2024. The closing price is set at 23:59 GMT.

The price sample is divided into two periods, with a pair identification and optimization period from August 1, 2021 to July 31, 2023, followed by a backtesting and validation period from August 1, 2023 to January 31, 2024.

3.2. *Pairs selection*

To select the relevant pairs for the implementation of pair-trading strategies, we adopt an iterative methodology described in figure 1.

3.2.1. Cointegration framework. In economics and finance, the majority of time series are non-stationary, giving rise to the spurious regression problems identified by Granger and Newbold (1974). A non-stationary time series exhibits trends, cycles, seasonal changes or other forms of structural variation.

Aware of this phenomenon, Granger (1981) introduced the theory of cointegration. Two non-stationary series are cointegrated if there is a linear combination of the two that is stationary. From an economic point of view, the concept of cointegration is akin to a long-term relationship in which it is possible to temporarily deviate from equilibrium, similar to short-term fluctuations.

All the econometric tests carried out in this study are at the 1% threshold.

To identify cointegrating relationships within our data sample, we must first ensure that all series are integrated according to the same order.† To this end, we perform stationarity tests to ensure the robustness of our results.‡ We remove 46 assets from our sample.§

We could have departed from the theory of cointegration by considering that two stationary assets can have a linear combination and influence investment strategies. However, for the sake of rigor and performance, we prefer to focus on the strict framework of cointegration. We make the assumption that a linear combination between two series that are already stationary would have a less violent mechanism for returning to equilibrium, making its residual similar to a white noise behavior.

To test cointegration between the remaining 162 assets, we use the Engle and Granger procedure. This involves regressing two series linearly against each other using ordinary least squares (OLS), and isolating the residuals from the regression:

$$P_{1,t} = \alpha + \beta P_{2,t} + \varepsilon_t \tag{1}$$

As the price series are left raw here, the constant, defined by α in the equation above, allows us to capture the difference in price scale between two assets and avoid bias in the analysis.

The stationarity of the regression residuals ($\hat{\varepsilon}_t$) is then assessed using the Dickey-Fuller test adapted with the tables of Phillips and Ouliaris (1990). Of the 13,041 possible pairs, the Engle and Granger test reveals 1,686 cointegrated pairs at the 1% threshold.

To put the results into perspective, we compare them with the less restrictive Johansen (1988)¶ procedure, which yields 3,773 pairs. By comparing these two procedures, upon verification we retain 1,541 pairs for the rest of our analysis (i.e there are linear combinations between the prices of assets in the cryptocurrency market, cf Hall and Jasiak 2024).

3.2.2. Construction of an error correction model. Engle and Granger (1987) representation theorem has shown that each cointegrated series can be represented by an error-correction model (ECM).‖ By setting up an ECM for each asset in the cointegrated pair, short- and long-term dynamics can be analyzed jointly, in order to reproduce the dynamics of adjustment towards long-term equilibrium. The two ECMs

¶The list of all assets is available in Supplemental material - Assets considered in the paper. Dataset available on https://data.mendeley.com/datasets/8pcf3vkmsv/1

† Series become stationary after being differentiated the same number of times.
‡ The augmented Dickey and Fuller (1979) test reveals 41 stationary assets, while the Phillips and Perron (1988) test reveals 11. The Zivot and Andrews (2002) test, meanwhile, identifies 139 price series as stationary, with a structural break at an unknown date. Finally, the Kwiatkowski *et al.* (1992) test appears to be the most rigorous, since no series is deemed to be stationary.
§ The assets removed from the analysis are available in Supplemental material - Stationary assets.
¶ Johansen's procedure is based on maximum likelihood estimation of a VECM.
‖ Granger's representation theorem also shows that the reciprocal of this relationship is true: if the series can be generated by an error-correction model, then they are cointegrated.

used for each cointegrated pair are as follows:

$$\Delta P_{1,t} = \alpha_1 - \lambda_1(P_{1,t-1} - \hat{\alpha} - \hat{\beta} P_{2,t-1}) + \varepsilon_{1,t} \quad (2)$$

$$\Delta P_{2,t} = \alpha_2 + \lambda_2(P_{1,t-1} - \hat{\alpha} - \hat{\beta} P_{2,t-1}) + \varepsilon_{2,t} \quad (3)$$

These models describe the variation of each series around its long-term trend. The choice of a simple model, i.e. with only one constant and one restoring force towards long-term equilibrium, is linked to the purpose of our modeling: the aim, rather than to try to predict the short-term dynamics of assets† is to explain the mechanisms that govern a pair as it returns to equilibrium.

Parameters λ_1 and λ_2 give an indication of the speed of each pair's adjustment towards its equilibrium level. If λ_1 is significant and negative for the first ECM, the first asset in the pair is participating in the return to long-term equilibrium of the cointegrating relationship linking the pair. Conversely, if λ_2 is significant and positive for the second ECM, then the pair's second asset is participating in the return to equilibrium. If the adjustment parameter, λ, is not significant for an asset, this indicates that it is not participating in the return-to-equilibrium mechanism: its evolution is independent of the value of the residuals in the long-term equation, and it satisfies the weak exogeneity criterion. In other words, this phenomenon means that the other asset in the pair, whose parameter is significant, alone ensures the existence of the cointegration relationship, thanks to its 'follower' price dynamics. The asset meeting the weak exogeneity criterion can be described as the pair's 'leading' asset.

In order to analyze the dynamics of adjustment towards long-term equilibrium, we build an ECM for each asset of the 1541 pairs selected. The observed results provide insights into the crypto-asset market in general. For the majority of the pairs studied, i.e. 1127 pairs, only one of the two assets participates in the return to equilibrium. Within these pairs, there are market leaders who cause other assets to oscillate around their trend. It is then possible to predict the short-term behavior of a pair's 'follower' asset in relation to the behavior of the 'leader' asset.

For 391 pairs, both assets participate in the return to equilibrium. In our study, it is these pairs that will interest us, as they are more conducive to the implementation of a pair-trading strategy. During a temporary deviation from long-term equilibrium, the price movements of these two assets will, in theory, allow their cointegrating relationship to return to equilibrium; in other words, positions taken as part of a pair-trading strategy will be more likely to result in gains with this configuration.‡

Finally, for 23 pairs, neither of the two assets participates in the return to equilibrium. This result, although not very intuitive at first glance, invites us to look for hidden variables that contribute to the return mechanism of these pairs, such as staking performance or the influence of other assets. If this research leads to the involvement of other crypto-asset prices, pair-trading strategies with more than two assets should be investigated, in particular looking for multiple cointegration relationships with Johansen's procedure. However, to date, no study suggests that multivariate cointegration is more interesting to trade than bivariate cointegration, especially as it would be difficult to find a common multiple and manage positions.

3.2.3. Evaluation of the long-term relationship residuals autocorrelation. To build pair-trading strategies, residuals of the cointegrating relationship, used to issue signals for opening and closing positions, must be exploitable. If the residuals are close to white noise, implementing a strategy around them is difficult because of the way they appear.§

In our study, we want the residuals to be autocorrelated in order to avoid trades that are too short and ultimately unprofitable. The typical behavior of the residuals is cyclical, so that they oscillate slowly around their mean (i.e. 0). To select only these pairs, we analyze the autocorrelation of the residuals by estimating AR(p) models. An AR(p) model with a relevant threshold could reflect a gradual adjustment of the residuals towards their central value.

An AR(p) model is built for the residuals of all 391 pairs. The parameter p of each model is chosen by maximizing the AIC criterion. To compare and remove pairs for which the residuals of their relationship do not behave in a way conducive to the desired strategies, we isolate the RMSE criterion of each AR(p) model.¶ Pairs which do not meet our threshold RMSE criterion (i.e. above average) are removed from our study, leaving 322 pairs that are retained.

3.2.4. Jumps filter. By looking at the residual series graphically, the presence of outliers, due to the non-normality of residuals,‖ is observable. These peaks can be interpreted as sudden breaks in the cointegration relationship between two assets. They appear, for the most part, during periods of turbulence on the crypto-asset market (i.e. in 2021, see figure 2).

We hypothesize that if a cointegrating relationship has experienced certain breaks in the past, it is likely to experience turbulence in the future. This type of relationship is dangerous in the context of a pair-trading strategy, since with the introduction of a stop-loss, an outlier can lead to heavy losses for the investor and high volatility. To avoid these relationships in our final sample, we apply a filter to all pairs. For each pair, we calculate the standard deviation of the residual series. If the series exceeds its standard deviation multiplied by five

† Huang and Martin (2019) have shown in this regard the limited usefulness of a richer error-correction model, of the ECM-GARCH type, to better anticipate the pair's behavior in the vicinity of extreme points and therefore to optimize the implementation of the initialization moments of buy or sell strategies on pairs.

‡ The intensity of the restoring forces appears quite low for the majority of pairs, with a median of significant restoring forces at 0.08. It would be interesting to compare this intensity with the speed of return to equilibrium for each pair.

§ i.e. residuals revolve around the average very quickly, making it impossible to be profitable and take viable positions over several days.

¶ The RMSE criterion is chosen here to compare the models, despite its non-standardization, because the models concern the residuals of the cointegrating relationships. Due to the presence of the constant in the linear relationship between the two assets, these residuals are of the same order of magnitude across all pairs.

‖ The Shapiro-Wilk test on the series of pairwise residuals showed that none of them was normal.

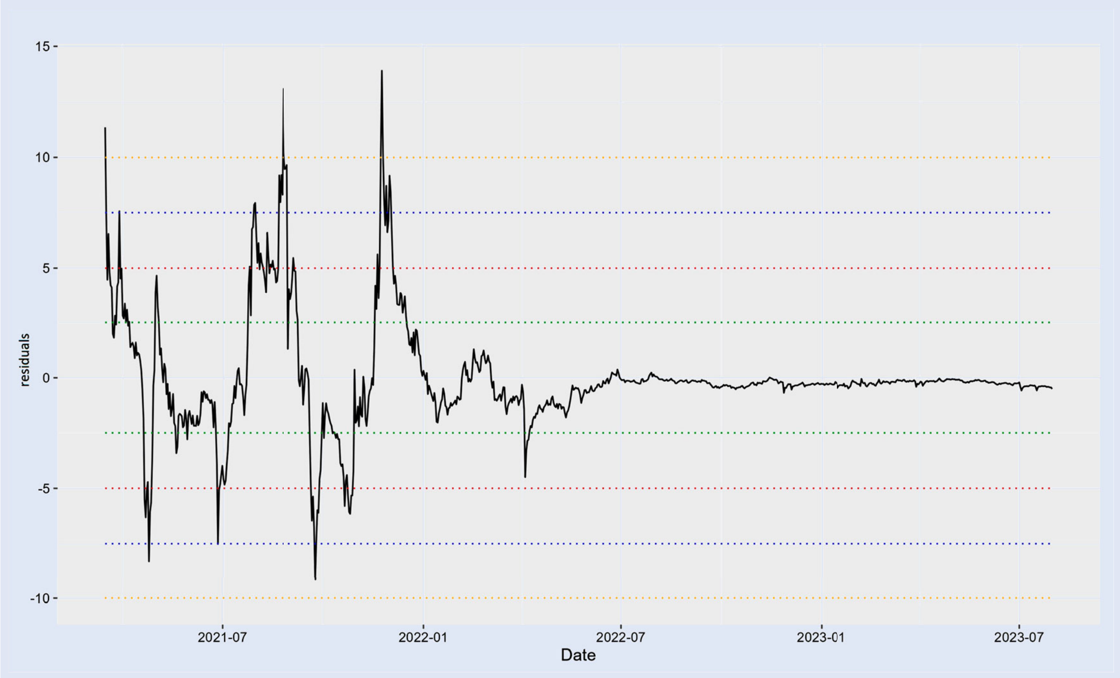


Figure 2. Outliers in residuals - ALICE CELO pair.

more than once, the couple is removed from our study. In this way, 93 pairs are excluded.

Our final sample consists of 229 pairs.† It is interesting to note that 74 assets present after the stationarity tests are not found in the selected pairs‡ (see figure 3).

This is due to one of two factors: either because assets are not cointegrated with others, in which case it would be relevant to analyze them as safe havens; or because the cointegrated pairs in which these assets are present did not pass the various filters applied, due to their volatility or their position as market 'leader' or 'follower'.

On average, one asset is present in five selected pairs. The assets most cointegrated with the others are not the biggest on the market, or the most well-known: the lack of a priori knowledge of potential relationships at the outset allows us to perceive relationships that would not correspond to market beliefs. The fact that the sector's dominant crypto-assets do not feature in the selected pairs is also explained by the choice to select only those pairs where the assets complement each other.

If we focus on the four largest crypto-assets in terms of capitalization (BTC, ETH, BNB and SOL): Bitcoin and Binance coin are not cointegrated with any of the selected assets. This lack of cointegration with the other assets in the industry begs the question: do these two major crypto-assets act as a safe-haven for the others§ ?

As for Ether, three assets are cointegrated with it (NULS, SFP and STPT). ECMs confirm our initial intuition: Ether is a market-leading asset. This fact does not surprise us, since the three assets cointegrated with Ether have a direct link with the Ethereum blockchain, since they exist as ERC-20 tokens.

In the final sample, Solana appears in only one pair, SFP/SOL, while it is cointegrated with six assets (AR, ATA, COTI, EGLD, IOTX and SFP). Dedicated ECMs show that for four pairs (ATA, COTI, EGLD and IOTX) only one of the two assets present participates in the return to equilibrium. Although these four crypto-assets have no direct fundamental links with Solana, the latter's importance in the industry can lead assets to oscillate around its trend, declaring it a 'leader'. The AR/SOL pair is excluded from the final sample, as its relationship is close to white noise behavior.

Finally, the asset most present in pairs is Firo, followed by Dexe. These two crypto-assets have different areas of focus. Firo, formerly known as Zcoin, is a cryptocurrency focused on privacy and security. It operates on a decentralized network with masternodes and a Proof of Work algorithm. In contrast, Dexe is the native token of the eponymous decentralized crypto-asset trading platform. It is primarily used for fees, platform governance and rewards.

3.3. *Strategy construction*

In a pair-trading strategy, a signal (buy or sell) on the pair is sent when the spread, or the residual of the relationship, between the two assets deviates from its mean. When this happens, inverse positions are opened on the two assets making up the pair. To take these two positions simultaneously, we

† All these pairs, and the assets that make them up, are available in Supplemental material - Selected pairs.

‡ The 229 pairs in the final sample are made up of 89 unique assets. These assets are present in Supplemental material - Assets present on the selected pairs.

§ Despite the absence of cointegration between Bitcoin and other assets, research suggests that the returns of Bitcoin may still exert a significant influence on the returns of other cryptocurrencies. This phenomenon can be attributed to Bitcoin's capacity to assimilate and reflect market information more rapidly than its counterparts, as demonstrated in the study conducted by Guo *et al.* (2024). The speed at which Bitcoin incorporates information into its price dynamics may account for the observed discrepancies and the lack of identifiable cointegration relationships with other cryptocurrencies.

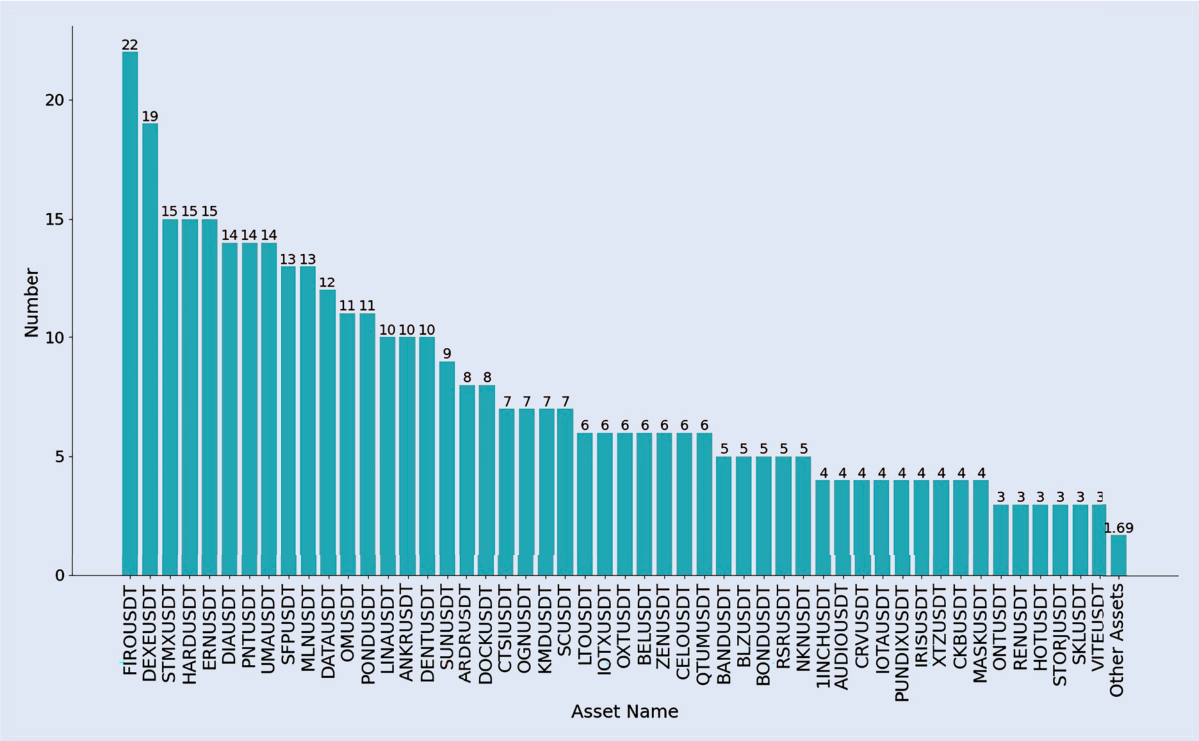


Figure 3. Assets involved in pairs.

exploit leverage within an investment. To respect the market-neutral aspect, these positions must take into account the cointegration coefficient, defined in the cointegration relationship. In addition, to take the short position on one of the two assets in the pair, we use leverage, in the same way as the SRD in traditional finance.¶

For a long signal on the pair, i.e. a buy on the first asset and a sell on the second, we can reason about the valuation of an elementary position (with a single security held as an asset in the position):

$$V_{p,t} = P_{1,t} - \beta P_{2,t} \tag{4}$$

The purchase of a security from the first asset (i.e. the asset side of the position's balance sheet) is offset by the short sale of securities from the second asset (i.e. the liability side of the position's balance sheet). We find the coefficient of the second asset on the linear regression of cointegration relationship, β, which is defined as the cointegration coefficient of the pair.

By generalizing the equation to n securities purchased on the first asset:

$$V_{p,t} = nP_{1,t} - n\beta P_{2,t} \tag{5}$$

$n\beta$ is the number of securities sold short on the second asset in order to capitalize on a bullish forecast of a return to the mean of the linear combination between the two assets (ε_t).

Let Ptf_t denote the amount of our portfolio at any time t, particularly at the beginning of a trade. According to our strategy, this entire portfolio is positioned on the long position of the pair (i.e. the purchase of securities of the first asset), the number of securities purchased is then:

$$n = \frac{Ptf_t}{P_{1,t}} \tag{6}$$

In turn, the number of shares in the second asset to be sold short is:

$$n\beta = \frac{Ptf_t}{P_{1,t}}\beta \tag{7}$$

Bringing the amount of the short position at the start of the trade to:

$$\frac{Ptf_t}{P_{1,t}}\beta P_{2,t} \tag{8}$$

For a short signal on the pair, i.e. a short sale on the first asset and a purchase of the second, the method is similar. The direction in which the equations are written is simply reversed, with n denoting the number of shares of the first asset to be sold short.

Thus, the valuation of the position generalized to n securities put on the short-sale position of the first asset, on the Liabilities side, is:

$$V_{p,t} = -nP_{1,t} + n\beta P_{2,t} \tag{9}$$

The amount of the portfolio is allocated to the long position of the trade, here on the second asset of the pair:

$$Ptf_t = n\beta P_{2,t} \tag{10}$$

The number of shares in the first asset to be sold short is therefore:

$$n = \frac{Ptf_t}{\beta P_{2,t}} \tag{11}$$

¶ This assumption takes into account unlimited collateral, as the long position is used as collateral to issue a short position on the other asset in the pair.

This brings the amount of the short position at the start of the trade to:

$$\frac{Ptf_t}{\beta P_{2,t}} P_{1,t} \tag{12}$$

The use of leverage in a pair-trading strategy is feasible and often beneficial for two reasons: (i) Leverage amplifies potential returns without significantly increasing net risk, as the positions are of a market-neutral type. (ii) Arbitraging market inefficiencies sometimes requires significant investment to make small price differences profitable. However, as pair-trading is a risky and potentially volatile strategy, a leverage limit of 2 has been introduced to moderate our exposure to risk. To implement this limit, we assume that, when there is a signal on a pair, the sum of the two positions must not exceed twice the portfolio valuation:

$$\max(long_t + short_t) = 2Ptf_t \tag{13}$$

When we receive a signal to take a position on a pair, we proceed in two steps. (i) We calculate the theoretical positions to be allocated to each asset according to the equations above. (ii) We check if the sum of the two positions does not exceed twice the amount in the portfolio at that moment. If this is the case, the calculated positions are implemented. If, on the other hand, the positions exceed the calculated amount, we use another set of equations to calculate a new amount to put on the long position in order to respect the leverage limit.

When there is a long signal on the pair, the amount to be sold short is:

$$short_t = \frac{long_t}{P_{1,t}} \beta P_{2,t} \tag{14}$$

By introducing this measure into the leverage limit equation:

$$long_t + \frac{long_t}{P_{1,t}} \beta P_{2,t} = 2Ptf_t \tag{15}$$

$$long_t \left(1 + \beta \frac{1}{P_{1_t}} P_{2,t}\right) = 2Ptf_t \tag{16}$$

By isolating the value of the long position, we can finally recalculate it, which will be less than the value of the portfolio (i.e. there is still some uninvested cash in our portfolio):

$$long_t = \frac{2Ptf_t}{1 + \beta \frac{1}{P_{1_t}} P_{2,t}} \tag{17}$$

The new amount of the short position is also calculated by repeating the equations with the revised amount of the long position.

In the event of a short signal on the pair, the principle is the same:

$$long_t + \frac{long_t}{P_{2,t}\beta} P_{1,t} = 2Ptf_t \tag{18}$$

By grouping the amount of the long position on one side, we obtain its new value:

$$long_t = \frac{2Ptf_t}{1 + \frac{1}{P_{2,t}\beta} P_{1,t}} \tag{19}$$

In the same way as for a long signal, the amount of the short position is revised according to the new calculated long position.

4. Results

4.1. Presentation of strategies

This paper is based on the comparison of four pair-trading strategies. Each of these strategies relies on the residuals of the relationship between the two assets to determine an opportune moment to buy or sell the pair. They are based on simple rules for calculating either fixed or dynamic standard deviations of the spread (or residuals of the cointegration relationship) that separates the two assets in a pair. Each of these strategies includes six levels: sell-entry, sell-stop-loss, sell-take-profit, buy-entry, buy-stop-loss, and buy-take-profit. For the sake of simplicity, we assume symmetry in the position-taking for both buying (i.e. buying the first asset and selling the second) and selling (selling the first asset and buying the second) of the pair. Therefore, we will only consider three thresholds going forward: the entry threshold, the take-profit threshold, and the stop-loss threshold.

The first strategy is the fixed strategy. It uses thresholds based on long-term calculated standard deviations to define the entry and exit points of a trade. For this, it relies on a fixed standard deviation calculated over the in-sample period. An example of this first strategy is illustrated in figure 4. The entry and exit thresholds for a trade remain the same throughout the entire period; they do not fluctuate. Thus, when the residual of the cointegration relationship crosses the entry level (yellow lines), a trade (either buying - green dots - or selling - red dots - the pair) is initiated. This trade is closed (orange dots) either when the relationship between the two assets experiences a significant deviation that reaches the stop-loss level (red lines) or when the relationship returns to normal (i.e. to its mean) and reaches the take-profit level (green lines). This strategy is the simplest one used, as it does not systematically recalculate the threshold levels and has straightforward rules.

The second strategy studied is the dynamic strategy. It is widely used in the literature and is named Bollinger Bands. Bollinger Bands are a tool invented by John Bollinger in the 1980s and have been a registered term since 2011. Derived from the concept of trading bands, they can be used to measure how ‘high’ or ‘low’ a price is in relation to past transactions. Bollinger Bands are made up of an N-period moving average MA, a band above K times an N-period standard deviation (SD) above the moving average (MA + K*SD) and a band below K times an N-period standard deviation below the moving average (MA - K*SD).

Standard deviation levels, unlike the fixed strategy, are no longer fixed but re-estimated for the last period N. The most commonly-used period N is 20 days. This standard configuration is recommended by Bollinger (2002) and adopted by traders for its balance between sensitivity to price movements and reduction of market noise. However, this period can be adjusted according to trading objectives and asset type. Day *et al.* (2023) found that a 60-day period brings better results when implementing the Bollinger Bands strategy on Bitcoin. Furthermore, in the context of a pair-trading strategy, a 60-day duration is relevant to achieve a balance between responsiveness and signal stability, especially when dealing with strategies involving correlated assets. Consequently, in our

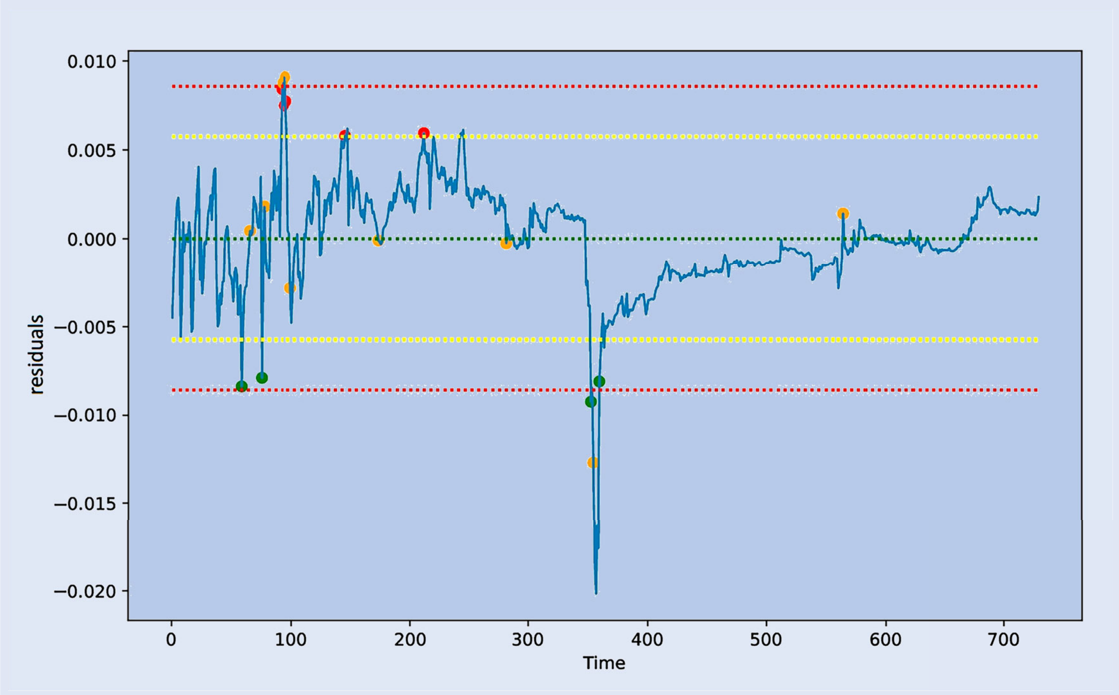


Figure 4. Residuals for the fixed strategy on a pair - AKRO BOND pair.

study, this period is set at 60 days. An example of this strategy applied to a residual of the cointegration relationship of one of the pairs is provided in figure 5. The only difference compared to the fixed strategy is the dynamic aspect in the calculation of the standard deviations of the residuals, resulting in a dynamic approach to position-taking on the pair.

The last two strategies introduce a rule in the initiation of a trade. Until now, the fixed strategy and the dynamic strategy result in simple decision rules: positions are opened if the residual is between the triggering threshold and the stop-loss level, and positions are closed if the residual exceeds the stop-loss level or reaches the take-profit threshold.

However, these simple rules do not account for a potential trend change in the cointegration relationship between two assets (i.e. a break). Although the dynamic strategy adapts through the rolling window for calculating the standard deviation, significant losses can be incurred during the readjustment period, as illustrated in figure 6.

To address this issue and to identify breaks or trend changes more quickly, a new rule concerning the initiation of trades is introduced in the third and fourth strategies.

The third strategy, referred to as the percentage strategy, consists in making a pair eligible for a new trade (after hitting a stop-loss) when the residual of its cointegrating relationship lies between the trigger threshold and this same threshold increased by a certain percentage. An example is provided in figure 7. In other words, with a percentage equal to 50%, if the pair's previous trade ends with a stop-loss, a new trade is opened (in the same direction) only if its residual lies between the trigger threshold and a limit situated 50% above this threshold (materialized by the yellow boundary in figure 7).

The fourth strategy, known as the return strategy, imposes a similar but more restrictive rule. When the last trade of a pair has ended in a stop-loss, the pair becomes eligible for a new trade only when its residual has touched or crossed the take-profit level. In other words, the return strategy requires a return to normalcy, i.e. to the mean, of its relationship in order to consider resuming a position on the pair. An example of this strategy is presented in figure 8.

4.2. Monitoring tool to anticipate cointegration breaks

Although introducing variants of the tested strategies can reduce the pairs portfolio's exposure to potential structural breaks by waiting for a return to normality before reinvesting, anticipating cointegration breaks in the out-of-sample period remains essential.

Accordingly, during backtesting we rely on a real-time monitor that computes the Phillips–Ouliaris cointegration test statistic on a daily basis. If this statistic decreases by more than 1% over the past three days,† the pair is abandoned (i.e. if it is currently in a trade the position is closed and the pair is no longer eligible for new trades until the end of the backtesting period) because it is considered potentially broken. An example of the evolution of the cointegration test statistic for the pair shown in figure 6 is provided in figure 9.

Since the test statistic accounts for all available past information, its response is lagged relative to the actual timing of a break. Nonetheless, thanks to this monitor we remain able to detect breaks and thus limit our losses during the backtesting period (see Supplementary Materials for further details).

† See sensitivity checks for these monitor parameter choices in the Supplemental material - Presentation and sensitivity checks of the real-time monitor.

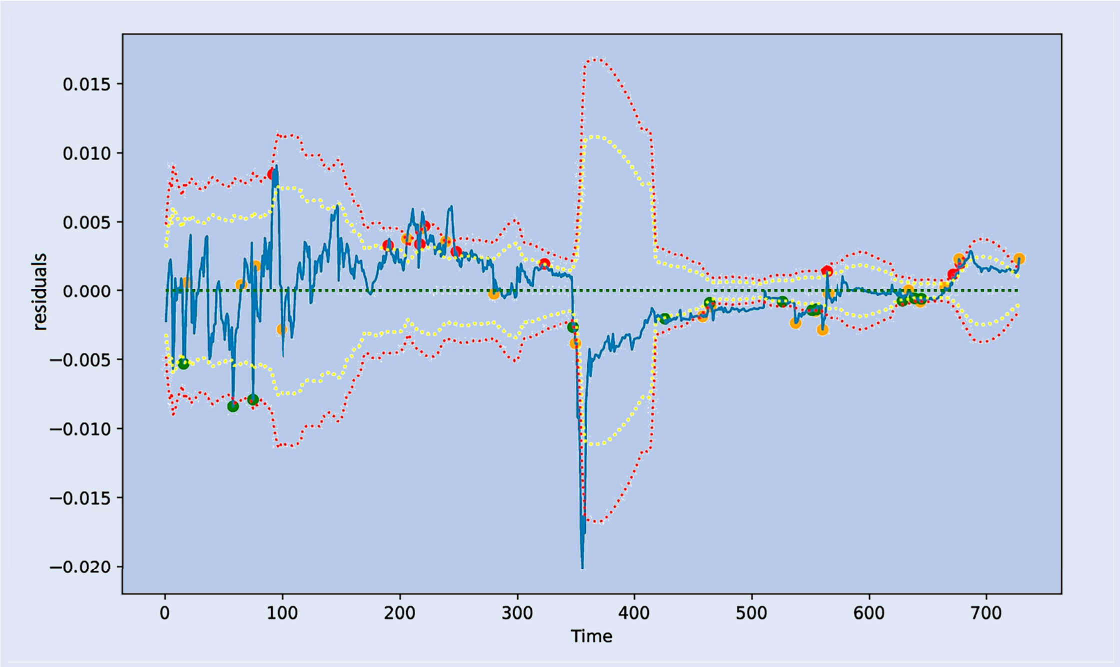


Figure 5. Example of Bollinger Bands residuals on a pair - AKRO BOND pair.

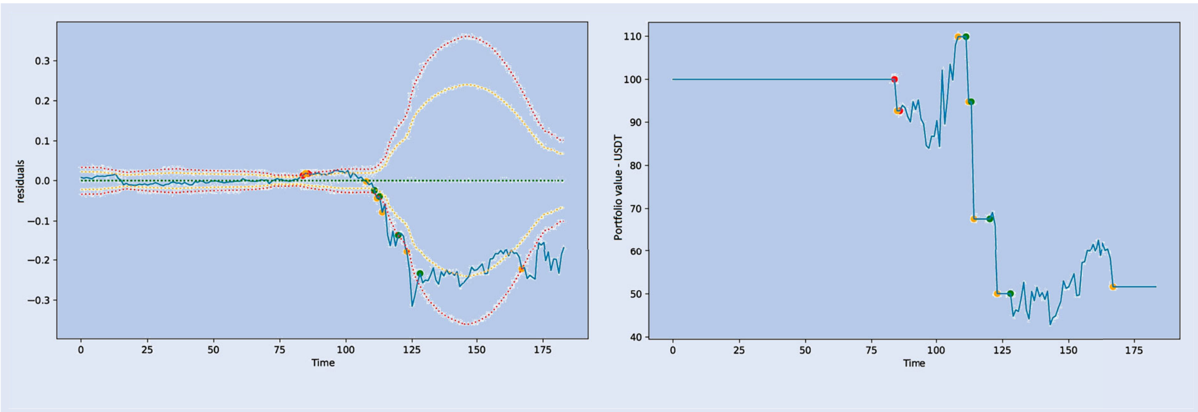


Figure 6. Example of the irrelevance of the identified stop-loss - CTSI SUPER pair. (a) Residuals over the backtest period and (b) Portfolio value in USDT over the backtest period.

4.3. Testing procedures for the strategies

For the calculation of positions and gains, we utilize the system of equations detailed in Section 3.3. To make the tests realistic, fees are charged for opening and closing positions. A fee of 0.1% is charged for entering and exiting a long position, and a fee of 0.5% for entering and exiting a short position. Apart from position taking, the portfolio is in the money market and provides a zero return.

Two periods are used to evaluate the strategies: the in-sample period is the period for optimizing the strategies, running from August 01, 2021 to July 31, 2023; the out-of sample period is used to verify the suitability of the chosen strategies, running from August 01, 2023 to January 31, 2024. At the start of each period, each pair's portfolio is made up of 100 USDT.†

To analyze the performance of the strategies studied, twelve indicators are calculated: cumulative return, annualized average daily return, standard deviation of returns (i.e. annualized average daily volatility), annualized Sharpe ratio,‡ max drawdown,§ return per trade, return per winning trade, loss per losing trade, number of trades, percentage of winning trades, percentage of stop-losses hit and average duration of a trade.

† USDT, also known as Tether, is a stable cryptocurrency (or stablecoin) that is indexed to the US dollar. This means that each USDT is designed to maintain a value equal to 1 USD. Stablecoins like USDT are used to provide price stability in the cryptocurrency universe, which is generally highly volatile. USDT is widely used on cryptocurrency exchange platforms to facilitate transactions, as it allows investors to transfer value without being exposed to the wide price fluctuations of other cryptocurrencies.

‡ The Sharpe ratio is calculated as the ratio between the annualized average daily return and the annualized volatility of returns. It is used to assess whether the additional return obtained by an investment is worth the risk taken.

§ Max drawdown measures the maximum loss an investor will incur if he buys at the top of the portfolio and sells at the next trough. This measure enables investors to understand the level of potential risk associated with a strategy and to which they are exposed.

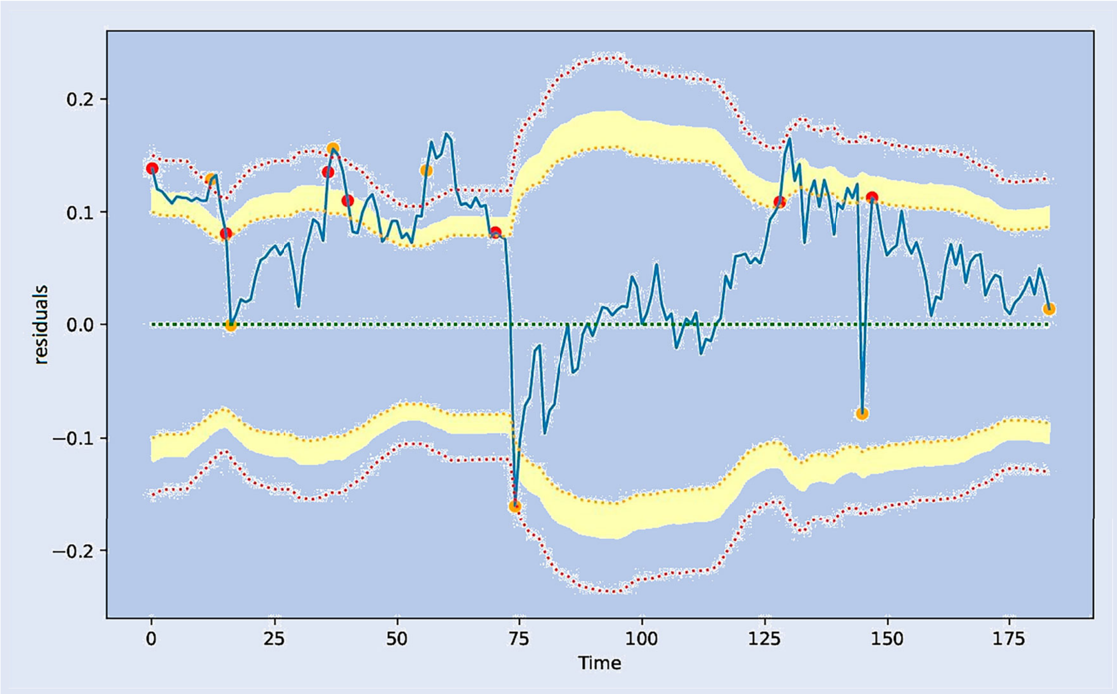


Figure 7. Cointegration residual applied to the percentage strategy - BEL BOND pair.

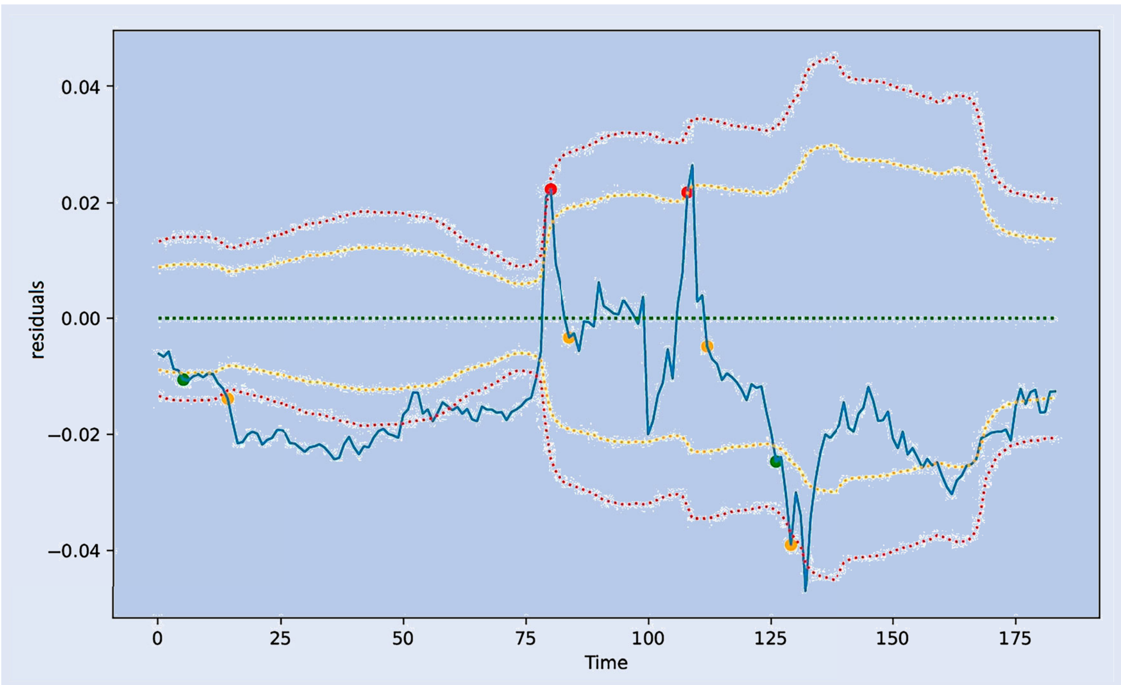


Figure 8. Cointegration residual applied to the return strategy - ARDR DEXE pair.

We analyze two types of tests: standard tests and optimized tests.

The standard tests set fixed levels in order to compare the four strategies against each other on a similar basis. Thus, the triggering thresholds are set at 2 standard deviations (for a sell order on the pair, by symmetry, the triggering threshold for a buy order is set at -2 standard deviations), the stop-loss levels at 3 standard deviations, and the take-profit level is defined as the mean of the residuals, which is 0. For the percentage strategy, the percentage level is set at 20%.

The optimized tests are conducted using a genetic algorithm. For each strategy, the genetic algorithm is defined to maximize the average Sharpe ratio of all the relevant pairs by optimizing the three (or four, for the percentage strategy) thresholds: the triggering threshold, the stop-loss threshold, the take-profit threshold, and the percentage level (only for the percentage strategy). Assuming symmetry in decision-making, the thresholds are expressed in absolute value but implicitly concern both buying and selling positions.

Figure 9. Evolution of Phillips-Ouliaris cointegration statistic - CTSI SUPER pair.

A genetic algorithm is an optimization method based on the principles of natural selection and genetics. It consists of several phases (see figure 10). The genesis phase randomly initializes a set of potential solutions (representing potential candidates for solving the problem). These potential candidates are evaluated according to their quality, or aptitude, to solve the given problem by maximizing a given criterion. The selection phase isolates the best-performing individuals (i.e. those optimizing the desired criterion) in order to reproduce and train a new generation. This is achieved via various reproductive techniques, such as crossover and mutation. Crossover involves combining the characteristics of parents to create offspring, while mutation introduces a small amount of random variation into individuals to maintain genetic diversity (so as not to identify only a local maximum). The new generation replaces the old one, and the steps described above are repeated with the new one. The algorithm stops when one of the termination conditions is reached, such as convergence towards an acceptable solution or exhaustion of the permitted number of iterations.

The genetic algorithm has an initial population of ten potential candidates, each candidate being defined as a vector of the three (or four, for the percentage strategy) parameters that we want to optimize, providing diversity in the selection of possibilities. It is defined to be iterated ten times. Known for its randomness, it is simulated several times to ensure the consistency of the results obtained.

Upon completion, the genetic algorithm provides a vector of three or four parameters that are optimized for pair-trading across our 229 pairs during the in-sample period. To define the boundaries within which the various parameters can vary, we rely on the literature. The maximum parameter bound (i.e. the maximum stop-loss level the strategy can adopt) takes into account the characteristics of the crypto-asset market. Taking into account a relevant stop-loss level is necessary for the strategy to enable the induced risk to be reduced while not cutting positions too early. The study by Białkowski (2020) thus helps to prove that introducing stop-losses into a strategy reduces the strategy's risk and sometimes its return for want of sufficiently precise calibration. The more volatile digital asset market requires adjustments to methods known from traditional markets for calculating optimal stop-loss levels. This work, requiring further research on the subject, could lead to an optimal stop-loss calculation method. The article by Kaminski and Lo (2014) shows that in momentum-type return-generating processes or regime-switching models, the introduction of a stop-loss rule can be beneficial in terms of return while reducing the induced risk. Although these articles do not deal with the specific case of pair-trading strategies, the principle of introducing a stop-loss is the same. In the event of a breakdown in the cointegrating relationship between the two assets, the introduction of an upstream stop-loss makes it possible to limit losses by exiting the position 'in time' (pending the response of the Phillips–Ouliaris cointegration test statistic – see monitor). Knowing that in Huang and Martin (2019) paper, the stop-loss level for their CFD pair-trading strategy is four, we decide to position our maximum bound at five. The bounds defined for the parameters are therefore [-1; 5]. The bounds for the percentage of trade eligibility in the first variant are [0; 100], with a constraint that the eligibility bound must not exceed the set stop-loss level.

Since the optimization is conducted over the in-sample period, this period is consequently overfitted. Therefore, we decide to focus on the backtesting period, or out-of-sample period, of our study. Although this period is relatively short (6 months), it allows us to understand and verify the validity of our study, particularly the role of a genetic algorithm in the decision-making process for opening positions. In figure 11, we compare the two types of tests across each of the strategies. The variable analyzed is the Sharpe ratio of each pair. It is noteworthy that for each strategy, optimizing the entry and exit levels of positions consistently achieves an average Sharpe ratio that is superior to that resulting from the implementation of strategies with predefined standard thresholds.

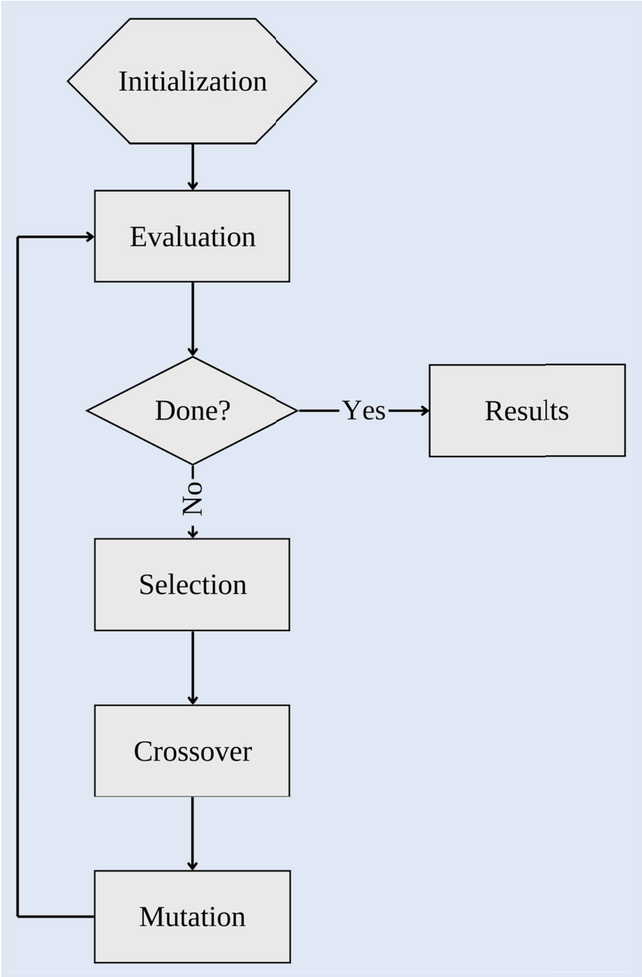


Figure 10. Genetic algorithm construction diagram.

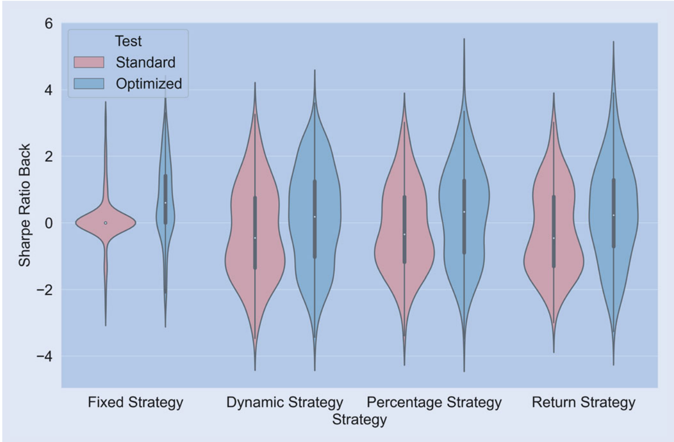


Figure 11. Comparison of Sharpe Ratio Between Standard and Optimized Tests for Each Strategy.

However, the distribution of the Sharpe ratios of the 229 pairs in the optimized tests appears more dispersed than that of the standard tests, primarily due to the optimized threshold levels being less constrained than those defined by the standard tests (i.e. the optimized stop-loss level is consistently close to the maximum boundary), leading to greater volatility in the strategy, which is not always compensated by the additional returns generated.†

4.4. Comparisons of the strategies

The comparison of the different strategies used involves numerous performance and risk indicators. Figure 12 summarizes the twelve indicators calculated for each strategy. The figures mentioned represent the average across the 229 pairs. The last column indicates the number of pairs that ended with a loss at the end of the backtesting period.

† Optimized thresholds [Trigger, Stop-Loss, Take-Profit]:

(i) Fixed Strategy: [1 ; 5 ; 1]
(ii) Dynamic Strategy: [1.56 ; 5 ; 0.24]
(iii) Percentage Strategy: [1.82 ; 5 ; 0.66], Percentage = 68.45
(iv) Return Strategy: [2 ; 5 ; 1]

As figure 11 indicated, the optimized tests yield better results. Indeed, although all strategies experience a significant increase in the average volatility of the pairs during the period (due to the optimized stop-loss threshold being consistently higher than the stop-loss threshold defined in the standard tests), the additional return generated more than compensates for the extra risk taken by the investor. For example, the Sharpe ratio of the return strategy is nearly two times greater.

Regarding the different strategies, it is noteworthy that the fixed strategy delivers the best risk-adjusted performance (with an average Sharpe ratio of 0.69 when optimized), even though it does not adapt to all pairs. Indeed, two distinct groups of pairs can be identified.

The first group consists of pairs that do not experience a change in amplitude and consistently reach the position triggering levels calculated from the long-term standard deviation (i.e. calculated over the in-sample period). These positions are mostly profitable (57.69% with the standard thresholds, 73.92% with the optimized thresholds), yielding the highest return per trade across all strategies (4.94% with the standard thresholds and 1.95% with the optimized thresholds). Considering only this group in the sample, the optimized fixed strategy delivers an average Sharpe ratio of 0.84. However, the number of pairs in this first group is relatively small – only 65 pairs (28% of the sample) have at least one trade under the standard fixed strategy. Under the optimized strategy, this number rises to 188 pairs (82% of the sample), due to the lower triggering threshold considered.

The second group concerns pairs that experience a change in amplitude in their long-term relationship. The year 2021 is known to have been highly disruptive in the cryptocurrency market, leading to the introduction of outliers in our sample. The largest outliers having been filtered out, most of our relationships experienced a significant decrease in their volatility after that year. However, the calculation of the standard deviation in the fixed strategy is based on the in-sample period, specifically from August 1, 2021, to July 31, 2023, and does not adapt to the market's fluctuations and trends thereafter. This example illustrates the rationale for testing the three other strategies mentioned, which use dynamic standard deviations and thus account for changes in the amplitude of the pairs' cointegration residuals.

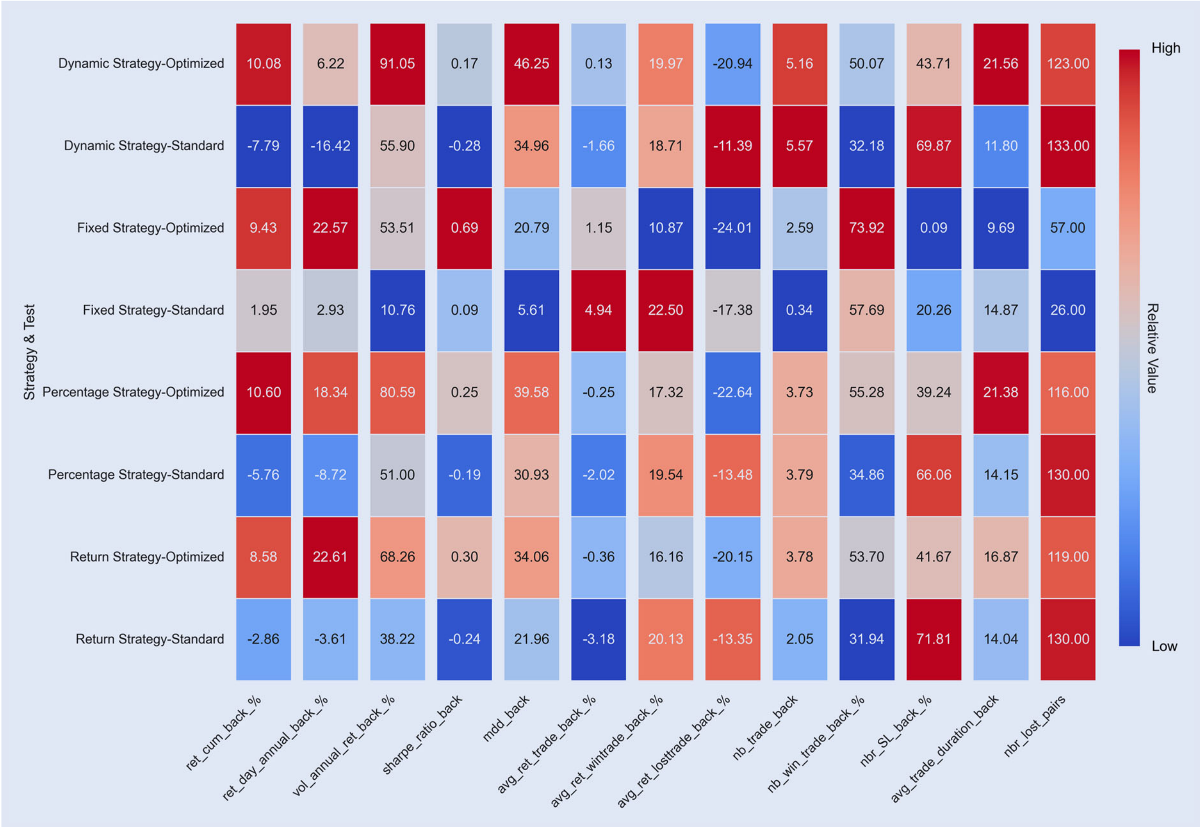


Figure 12. Heatmap of Key Strategy Indicators. (All columns are on their own scale).

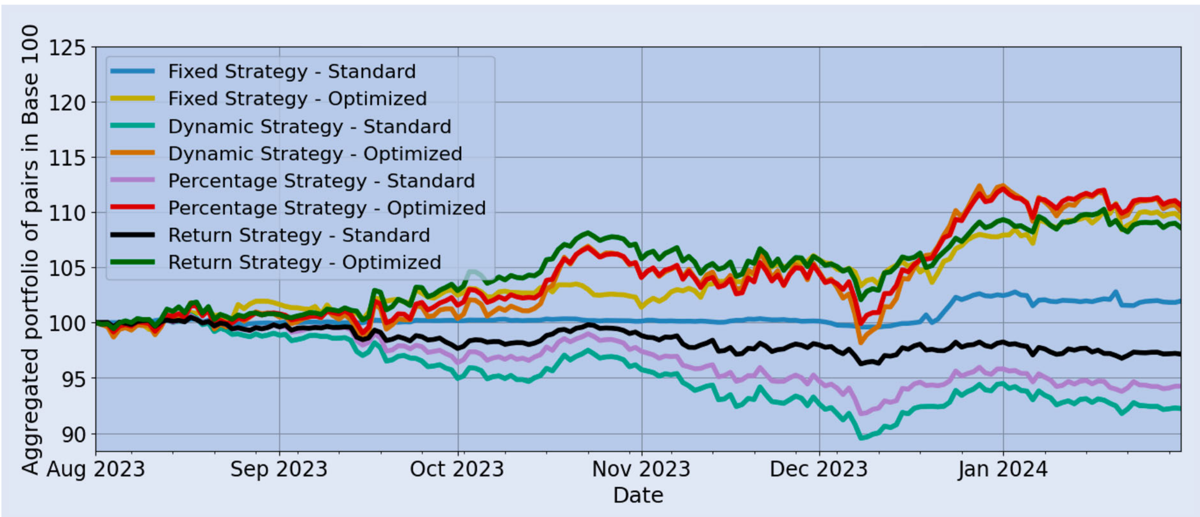


Figure 13. Performance of pairs aggregated in base 100 by strategies.

Across these three strategies, the results are mixed. At the standard thresholds considered, these strategies fail to deliver performance because the stop-loss threshold is too low (nearly 70% of trades end in a stop-loss). At the optimized thresholds, results are better, with an average cumulative performance higher than the fixed strategy (10.60% for the percentage strategy, 10.08% for the dynamic strategy) and a reduction in stop-loss hits of more than 20 percentage points compared with the standard thresholds (i.e. we also observe that the optimized stop-loss threshold is systematically at the upper bound of 5).

However, even with a higher cumulative performance, the additional volatility generated by these three strategies is proportionally greater than the gains achieved, reducing the average Sharpe ratio to 0.30.

Moreover, the proportion of losing pairs is significantly higher than with the fixed strategy. Considering only pairs that have at least one trade during the period, the dynamic, percentage and return strategies average between 50% and 57% losing pairs, compared with 30%–40% for the fixed strategy.

Ultimately, the application of pair-trading strategies in the cryptoasset sector appears interesting and relevant. While the

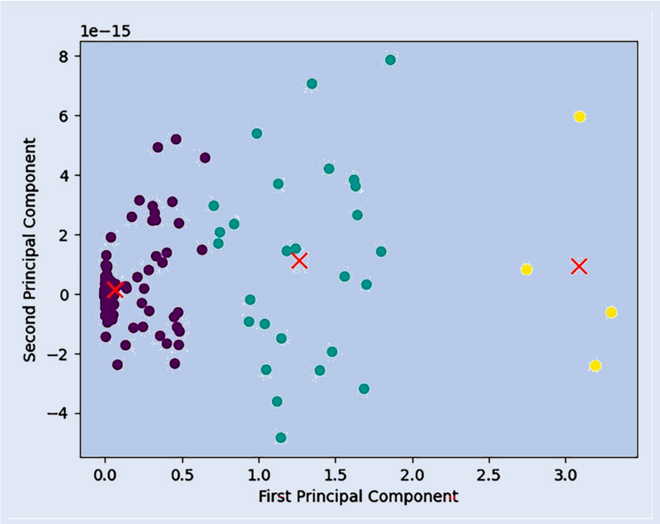


Figure 14. Pair clustering by volatility of cointegrating residuals.

volatility of these strategies remains very high, it would be worthwhile to explore constructing pair portfolios aimed at reducing it while maximizing profit. In figure 13, we present a graphical evolution of a portfolio based on 100 at the beginning of the backtesting period. The portfolios are constructed to be equally weighted across all pairs. All optimized strategies appear profitable by the end of the period, with lower volatility for the fixed strategy (notably in December 2023).

4.5. Comparisons with the literature

In relation to the literature, our paper aligns with the various results obtained in previous studies. Like Lesa and Hochreiter (2023), we find that pair trading applied to a basket of cryptocurrencies using cointegration as a selection method is effective and can deliver a positive return, even over a short period. This primary result reinforces the findings published by Rad *et al.* (2016), which demonstrate that cointegration outperforms other pair selection methods in turbulent market contexts.

Moreover, the use of a genetic algorithm to find the optimized thresholds for strategies proves to be a valuable tool, as the observed results suggest a systematic superiority of optimized strategies over standard strategies with thresholds defined in the literature. This is noteworthy as it supports the idea that genetic algorithms are relevant for strategy optimization and can replace simpler optimizers such as constrained optimization or Monte Carlo simulations (Zhang 2021). Indeed, in this context, the genetic algorithm appears more pertinent as it converges and iterates proposals based on the success of past proposals, unlike Monte Carlo simulations, which result in testing multiple propositions.

Finally, the analysis of pair trading across a broad basket of cryptocurrencies allows us to confirm that it is a volatile market and that relationships, even those identified as cointegrated, can be 'falsely' linked or subject to breaks in cointegration, as expressed by Fil and Kristoufek (2020). This connection is evident through the introduction of a stop-loss, which can have an extremely detrimental effect leading to underperformance. Here, we observe this concept as the optimized stop-loss levels are consistently close to the maximum allowed boundary, indicating either that the relationships are very volatile or that there may be recurrent short-term breaks in the relationships. Moreover, the monitor introduced to anticipate breaks in cointegration relationships during backtesting issues signals for 97 pairs (i.e. 42% of our sample).

5. Discussion, robustness and limitations

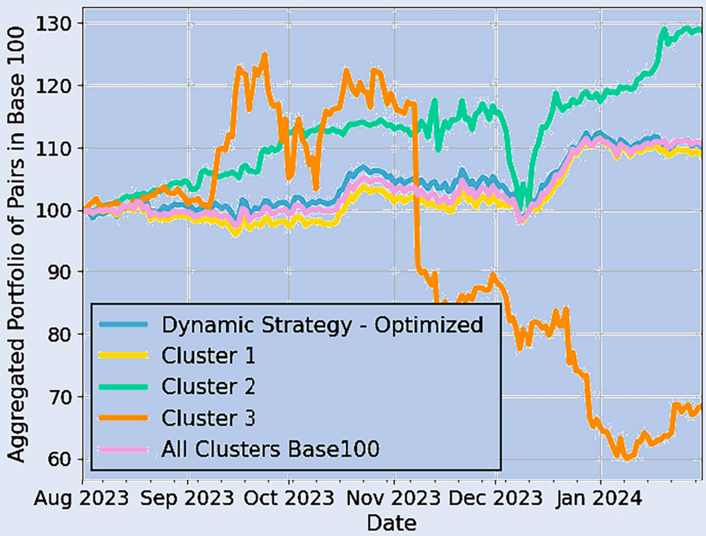
We would like to present a few additional thoughts and results. These concern the optimization of pair-trading strategies refined by risk class, the analysis of pairs in loss situations, the implications of our results for the efficiency of crypto-asset markets and finally some limitations with regard to our results.

5.1. Risk classes

The implementations of pair-trading strategies in the crypto-asset market are mitigated, between returns and risks involved. In an attempt to reduce risk and understand where performance comes from, we classify pairs according to the volatility of their residuals. Using K-means, three clusters are identified (see figure 14). The first cluster contains the majority of pairs, 198, with low volatility and stable relationships over time. The second cluster, comprising 27 pairs, brings together those with a slightly volatile relationship over time. Finally, the third cluster, made up of 4 pairs, contains pairs with highly turbulent long-term relationships. Modeling and optimizing pair-trading strategies according to these three clusters has shown that these strategies work best for pairs with a perennial cointegrating relationship over time. In view of these results, and from a portfolio construction perspective, it would be appropriate to adapt our sample of pairs and retain only those with a stable relationship.†

To compare the results of each cluster both individually and collectively, we use the dynamic strategy and apply it to the identified classes (see figure 15). The majority of pairs are in the first cluster; because of its greater diversity, it provides stability throughout the period and delivers positive return. This result supports our belief that classifying pairs via common characteristics, such as the volatility of their long-term relationship, enhances and captures all the alpha present in the pairs. The second cluster experiences greater fluctuations in trend, but delivers better performance, with a cumulative return at the end of the period of around 28%. The increase in volatility is offset by performance, since it is the cluster that achieves the highest average Sharpe ratio of 0.54. The final cluster, characterized by the most volatile relationships, undergoes numerous shocks in its trend, particularly from November 2023 onward. This cluster ultimately delivers a negative return over the period with a poor Sharpe ratio.

† For more information on the construction of the classification, please refer to Appendix 1 of the appendix.



Strategy	Mean Sharpe ratio
Dynamic Strategy	0.17
Cluster 1	0.2
Cluster 2	0.54
Cluster 3	-0.91
All Clusters	0.22

Figure 15. Aggregate performance of tested strategies over the backtest period.

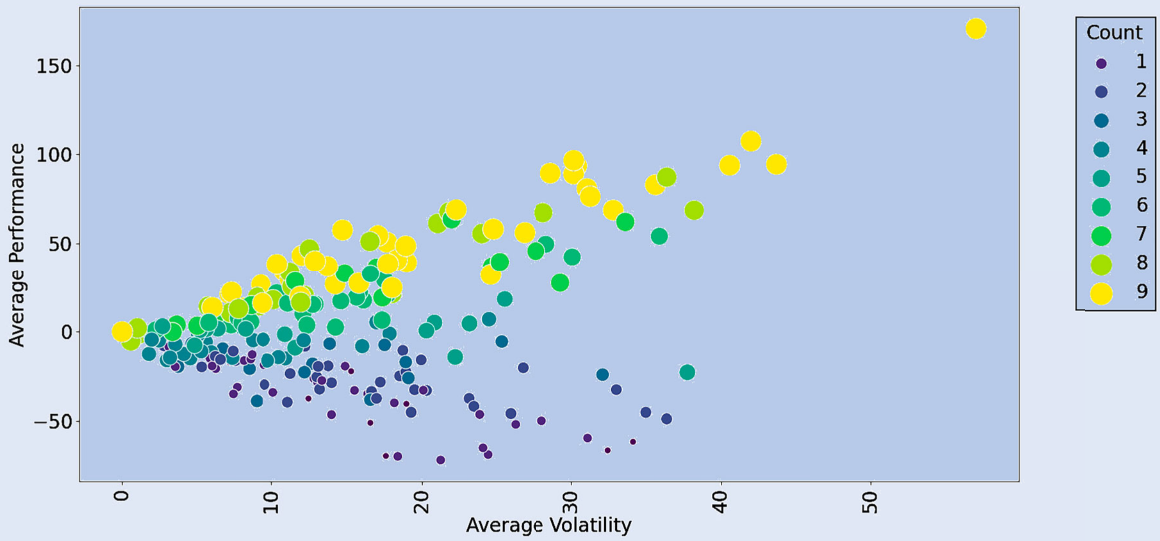


Figure 16. Bubble plot of pairs with winning strategies count and average performance/volatility.

Ultimately, although this classification and optimization maximize performance, the third class accentuate volatility in the strategy (with optimized thresholds which are more extensive than those of the other two classes‡).

5.2. *Pairs analysis*

Even though the strategies work in aggregate, there are disparities between the pairs selected. Of all the strategies tested, 189 pairs out of the 229 end up in loss in at least one of the tests carried out (see figure 16). The positive correlation between return and risk is confirmed here.

The out-of-sample monitor allows us to anticipate certain breaks in cointegration relationships. For example, four pairs† are analyzed, the particularity being that they all contain the same crypto-asset: OM (mantra). Mantra is the token of Mantra DAO, a decentralized platform on Ethereum that focuses on staking, lending and yield generation services. Its token, OM, is used for staking and governance.

Taking a closer look at the series of these five assets involved, see figure 17, by the end of 2023, the mantra/OM had dropped completely in level, without taking down with it the other assets with which it is usually cointegrated. This was due to governance decisions by the decentralized Mantra

‡ Optimized thresholds of clusters [Trigger, Stop-Loss, Take-Profit]:

(i) Cluster 1: [2.09 ; 5 ; 0]
(ii) Cluster 2: [2.92 ; 5 ; 0]
(iii) Cluster 3: [2 ; 4.26 ; -1]

† OM/STMX, OM/ZEN, OM/PNT and OM/REN

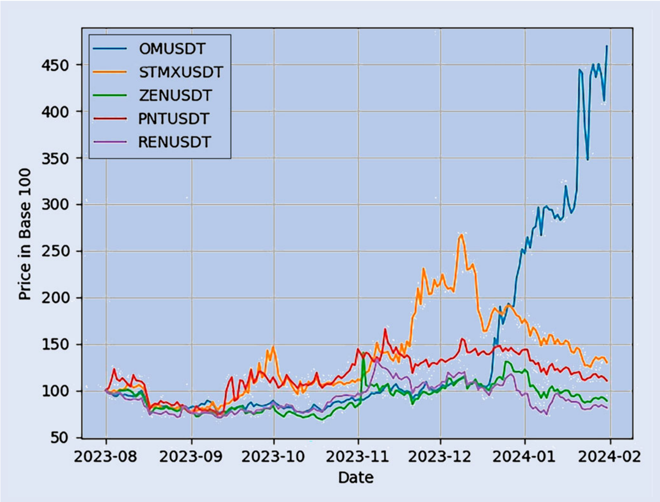


Figure 17. Price chart in base 100 of five assets involved in loss–making pairs.

DAO protocol, which decided on several occasions in January 2024 to significantly increase the staking yield assigned to the OM token.‡ As a result, more investors were attracted to the token and its potential staking return, driving up its price. Finally, the inflated price of OM led to breaks in these four known cointegrating relationships (see Appendix 2 in appendix). In this and some cases, the monitor effectively limits its losses resulting from breaks in cointegration relationships – the four pairs illustrated here manage to deliver profits under certain strategies. Conversely, in other situations the monitor responds too late: four pairs involving the asset SC - Siacoin - incur losses across all strategies§ .

5.3. *Implications for market efficiency*

Overall, our results reveal that, for a significant number of crypto-asset pairs, price forecasting - or more precisely, linear price combinations - becomes possible under particular market conditions, notably when cointegrating relationships deviate too far from their central value. Not only are such forecasts possible, they also form the basis of systematic trading strategies that prove profitable according to the usual financial metrics used in this paper.

This form of price predictability can be interpreted as evidence of temporary market inefficiencies, where prices deviate from a perceived fundamental level for the valuation differences between two crypto-assets, before gradually correcting the anomaly.

It is specifically the weak form of market efficiency that is being challenged here, albeit in a temporary manner.

The high volatility of prices and price differentials seems legitimate for digital assets that do not distribute income (setting aside the role of stacking for certain assets). The valuation of these assets is therefore not anchored in revenue expectations, as is the case for conventional financial assets (dividends, coupons, etc.). It is essentially speculative, based on confidence in the current and future uses of cryptocurrencies as payment and investment instruments. This refers not only to the technical characteristics of the various cryptocurrencies as payment instruments, but also to the reliability of trading platforms. Confidence in the development of the cryptocurrency ecosystem also plays a role in valuations, probably as a risk factor common to most digital assets.

We can conjecture that the standard deviations of the cointegration relationships partly reflect uncertainty about the fundamental values of the various cryptocurrencies, and even more precisely about the differences in fundamental values between them. There are, of course, other factors that can influence the volatility of prices and price spreads. These could be identified through the prism of behavioral finance. They include the irrationality of certain investors, mimicry, market manipulation (notably through the dissemination of information on social networks), and the fragmentation of trading platforms.

Overall, pair-trading strategies such as those proposed in our research are likely to smooth mitigate arbitrage opportunities and the associated pockets of inefficiency. However, they are unlikely to eliminate them completely, given that incomprehensible uncertainty still surrounds investors' perception of the fundamentals of crypto-asset prices and spreads. In other words, these strategies bolster investors' assessments of fundamental discrepancies between crypto-assets.

5.4. *Limitations*

As previously noted with losing pairs, the primary limitation of our pair-trading strategies lies in the potential breaks in the cointegration relationships between digital assets. Although the monitor, which computes the Phillips–Ouliaris cointegration test statistic on a daily basis, can detect and mitigate losses for certain pairs, it has limitations due to its slow response: by incorporating the full history of past information, it may react too late to recent breaks. Practitioners interested in implementing this type of strategy should therefore pay particular attention when using and interpreting this indicator to ensure the stability of the cointegration relationships they trade. Moreover, implementing stop-loss mechanisms remains an effective measure to guard against any atypical behavior of the pair out-of-sample and in trading conditions.

A second limitation concerns the liquidity of the assets involved in the pairs. This liquidity can be heterogeneous across assets and vary over. Furthermore, it may be impacted by pair-trading strategies such as those proposed in the article.

It should also be noted that not all the assets selected in our study are currently eligible for short selling.

Finally, the brokerage fees used in the study are fixed for all pairs and may thus differ marginally from actual brokerage practices.†

‡ https://x.com/MANTRA_Chain/status/1747921782361768391, https://x.com/MANTRA_Chain/status/1749325959889830373, https://x.com/MANTRA_Chain/status/1748337268014911558, https://x.com/mantra_chain/status/1750494405407236194 (accessed 1 June 2024)

§ DIA/SC, FIRO/SC, LINA/SC, QTUM/SC

† Sensitivity tests on fees showed that the deterioration in performance indicators was proportional to the increase in transaction costs, see Supplemental material - Sensitivity checks on fees.

6. Conclusion

This paper set out to study optimal pair-trading strategies within the formal framework of cointegration.

Its originality with respect to the existing literature on the subject lies in four points. First, our study is based on a very broad panel of digital assets. We make no assumptions about the potential relationships between assets. In total, we base our work on 208 assets listed on the Binance centralized exchange since August 1, 2021. Second, our study is based on a longer timeframe than the existing literature on the subject. Indeed, as this market is still in its infancy, previous studies have not had such a lengthy timeframe at their disposal. Another original aspect of our paper is the desire to understand the behavior of each asset within a pair. Thanks to the construction of a simple error-correction model, we are able to explain the mechanisms of return to equilibrium and to identify, or not, leaders within a pair. Finally, the choice of a genetic algorithm to optimize the entry and exit thresholds of our pair-trading strategies on the crypto-asset market is innovative, since we are able to combine classical econometrics for pairs selection and machine learning for strategy construction.

In terms of pairs selection, some of our results show that, for certain pairs, neither asset participates in the return to equilibrium. This fact invites us to look for other variables which contribute to the mechanism of return. There are two possibilities. If the other variables concern the prices of other crypto-assets, then it would make sense to examine pair-trading strategies with pairs made up of more than two assets. It is also conceivable that the return mechanism of these pairs obeys asset-specific variables such as staking returns. In this case, it would be interesting to integrate these variables into the error-correction model, in order to quantify their contribution to the return mechanism in the event of a divergence in the long-term relationship.

In terms of strategies, our initial results suggest that the strategy using historically calculated standard deviations delivers better outcomes than the strategies tested with dynamic 60-day rolling standard deviations. Indeed, with the standard thresholds applied, the stop-loss threshold (set at 3 standard deviations) appears too tight, causing the majority of trades executed under the dynamic standard deviation strategies to end by hitting a stop-loss (approximately 70%). Conversely, the fixed standard deviation strategy, although it does not capture all opportunities due to its failure to account for changes in the residuals' volatility for most pairs, manages to deliver a positive Sharpe ratio.

The optimization of parameters via a genetic algorithm significantly enhances results, yielding an average Sharpe ratio of 0.69 for the fixed strategy. Although the dynamic standard deviation strategies are able to produce positive performance, they are penalized by a proportionally higher induced risk relative to the gains generated, with annualized volatility of 91.05% for the dynamic strategy, 80.59% for the percentage strategy, and 68.26% for the return strategy.

The implementation of an automatic real-time monitor for detecting and anticipating breaks in cointegration relationships during backtesting allows us to reduce losses and make the solution practicable. Over the six-month period, the monitor issues an alert for 97 pairs (i.e. 42% of our sample).

At the end of the article, we try to go a step further by implementing a K-means classification of our pairs according to the volatility of their relationships. The results obtained are particularly interesting, since they highlight that a pair-trading strategy works best on pairs with a stable relationship. These are the pairs that generate the highest alpha in terms of risk/return ratio. For the future, it would be worthwhile to pay particular attention to these pairs when constructing a portfolio involving several pairs at different scales.

Finally, our article shows that cointegration relationships exist in the crypto-asset market. These relationships can be exploited as part of a pair-trading strategy, and deliver good results in terms of risk/return ratio, even though the volatility of these strategies remains relatively high compared to those implemented on traditional markets (see Appendix 3). Furthermore, the backtest used in this paper reflects the lower bound of outcomes attainable under an active strategy, such as regularly re-estimating the model. These results enable a comparison with the efficiency of the crypto-asset market. The existence of profitable pair-trading strategies confirms the existence of pockets of market inefficiency, characterized by short-term deviations of asset prices involved in the pairs from their equilibrium trajectory.

Future research will consist in cross-referencing all the results obtained on cointegration relationships, mechanisms for returning to equilibrium, the intensity of recall forces, and the performance of pair-trading strategies with the objective characteristics of crypto-assets: associated economic activities, outstanding amounts and liquidity of assets, the possibility and mechanism of short selling, the existence of stacking, etc.

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Data availability statement

The data that support the findings of this study are openly available in mendeley at https://data.mendeley.com/datasets/8pcf3vkmsv/1.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Supplemental data

Supplemental data for this article can be accessed online at https://doi.org/10.1080/14697688.2026.2653663.

7. Open Scholarship

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Appendices

Appendix 1. Classification of the pairs

From the graphical analyses we have carried out, we have noticed that the residuals of the cointegrating relationships of the selected

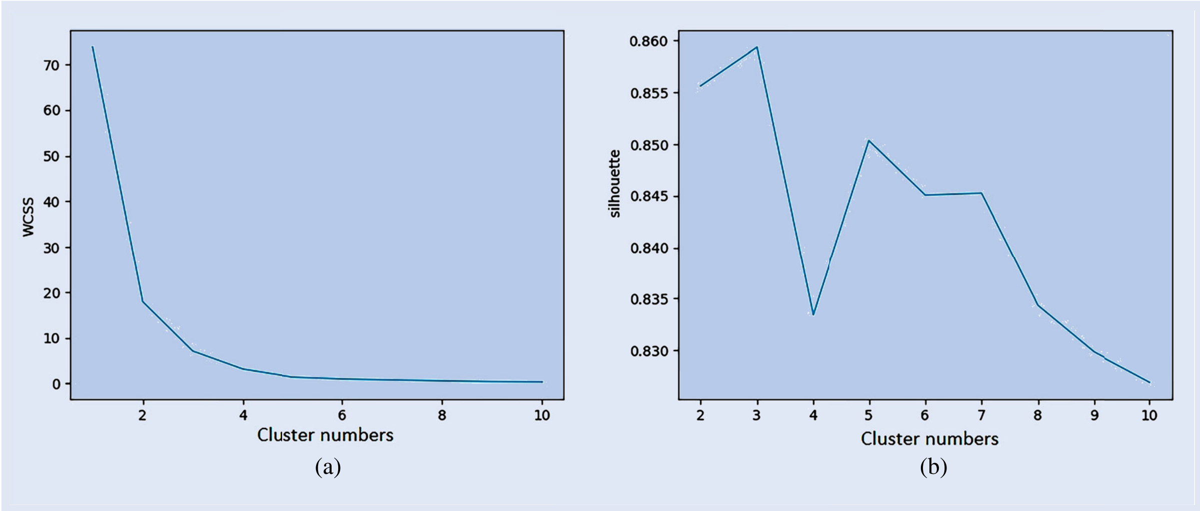

Figure A1. Method for selecting the number of clusters. (a) Elbow method and (b) Silhouette coefficient.

pairs do not all have the same volatility amplitude. Based on this postulate, we want to test whether a certain type of volatility amplitude is preferable when implementing pair-trading strategies or not. To do this, we want to classify pairs according to the volatility of their residuals. By classifying pairs into several categories, it might be possible to determine more suitable and efficient optimization parameters for each category. To cluster pairs according to the volatility of their residuals, we use k-means. K-means is often considered one of the best clustering algorithms due to its simplicity, efficiency and ability to handle large datasets. K-means is used to group individuals with similar characteristics (i.e. clusters) through the analysis of two defined components of the dataset. This unsupervised learning algorithm segments individuals into 'K' groups by minimizing the Euclidean distance between a given individual and the center of a cluster. K-means has certain limitations, the main one being the determination of the optimal position of the centroids of the initial clusters during the first iteration. These limitations have been addressed by implementing new approaches to reduce the number of iterations and the execution time, as in the paper by Zubair *et al.* (2022). Despite its limitations, k-means continues to be used in many fields, such as e-commerce with the recent paper by Prabhas *et al.* (2023), medicine with the paper by Kakushadze and Yu (2017) or finance with the paper, for example, by Dang *et al.* (2022). In fact, it enables individuals to be clustered, without apriori, by determining common patterns and without being programmed to predict a value from an analysis. Clustering places a function under the principle of exclusive membership. In other words, the same data cannot be found in two different clusters. We use two methods to determine the optimum number of clusters according to the mean of the residuals and the variance of the residuals in each pair. Correctly determining the number of clusters ensures that the data is divided efficiently and correctly. An appropriate value for this number 'K' helps to maintain a good balance between compressibility and precision.

The first is the elbow method. It is based on the fact that the sum of intra-cluster variance can be reduced by increasing the number of clusters. The higher the number of clusters, the more refined groups can be extracted from the analysis of data objects that are more similar to each other. We use the turning point of the sum-of-variances curve to select the right number of clusters. This method applied to our data is shown in figure A1(a).

The second method used is the silhouette score. This method evaluates the quality of the clusters created by the clustering algorithms. Ranging from $[-1, 1]$, the silhouette score is sometimes used to find the optimal value for the number of clusters 'K'. To do this, we consider the value of 'K' with the silhouette score closest to 1. The silhouette method applied to our pairs is shown in figure A1(b).

Unanimously, both methods obtain the optimal value of three clusters ('K' = 3).

We therefore parameterize the K-means to make three clusters, and the result is shown in figure 13. Three clusters can be distinguished according to the variance of the pairs' cointegrating residuals.

The first and largest cluster groups 198 pairs (86.46% of the sample). The characteristic feature of this cluster is the low volatility (variance less than 0.5) of the cointegrating residuals of each pair.

The second cluster, made up of 27 pairs, has the characteristic of grouping together pairs with average residual variance, between 0.5 and 2.

Finally, the last cluster groups together 4 pairs with a high variance of cointegrating residuals (between 2.5 and 3.5). These four pairs can be seen here as outliers.

Appendix 2. Residuals of losing pairs

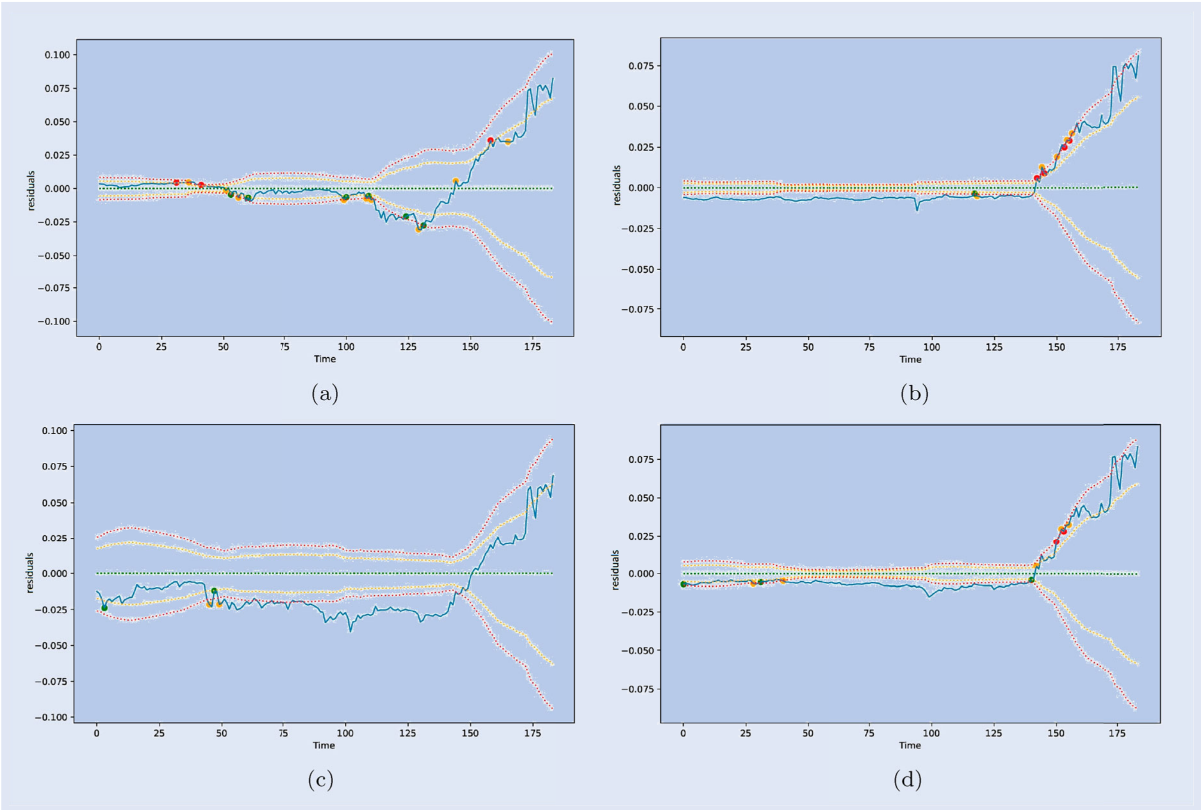

Figure A1. Residuals of pairs with break in cointegration relationship over the backtest period. (a) OM PNT. (b) OM REN. (c) OM STMX and (d) OM ZEN.

Appendix 3. Pairs example

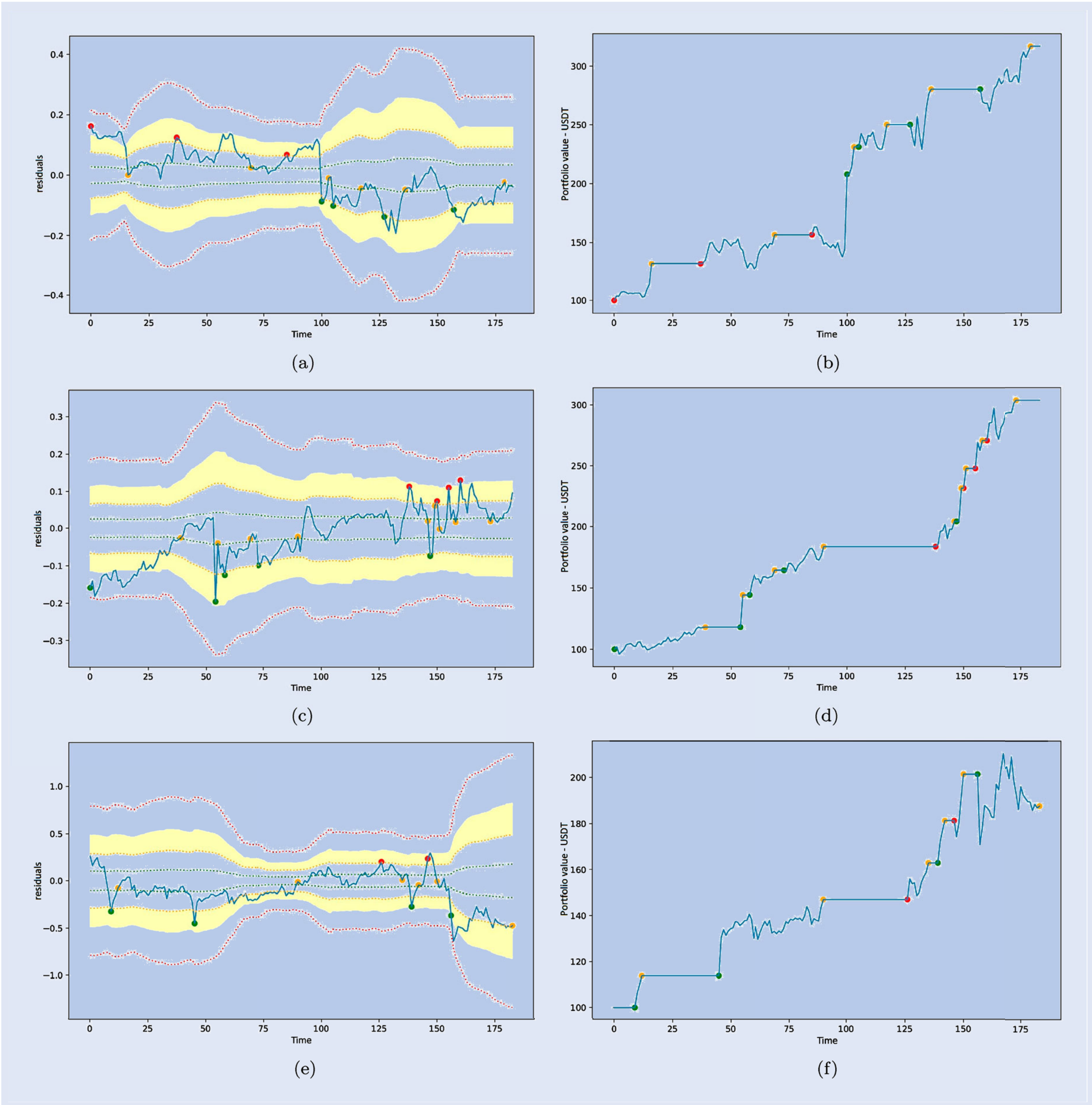

Figure A2. Example of pairs identified in our article in the out-of-sample period. (a) BEL DEXE - Residuals. (b) BEL DEXE - Portfolio value. (c) CELO MLN - Residuals. (d) CELO MLN - Portfolio value. (e) FIRO OXT - Residuals and (f) FIRO OXT - Portfolio value.